

Design of Voltage Reference Circuit

Vinayak Hande

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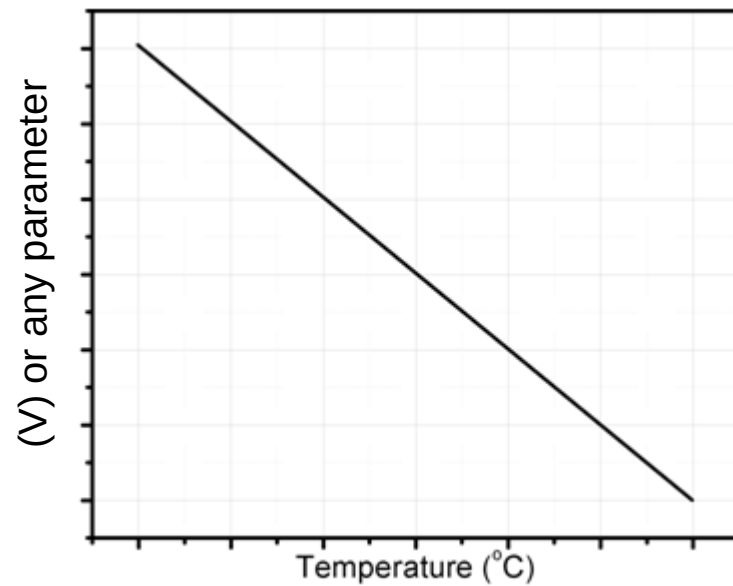


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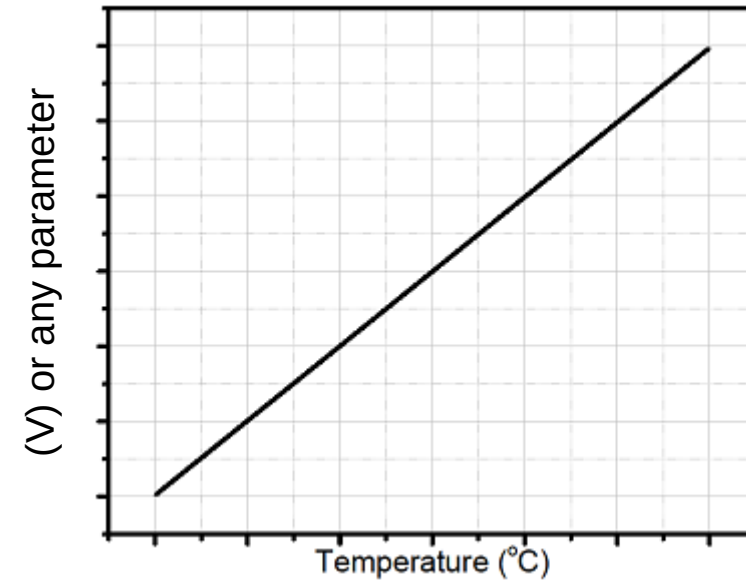
1. Basics of voltage references
2. Popular topologies of voltage references
3. Feedback Analysis
4. Case study
5. Conclusion : Part 1

Terminologies

(1) CTAT (Complementary to absolute temperature)

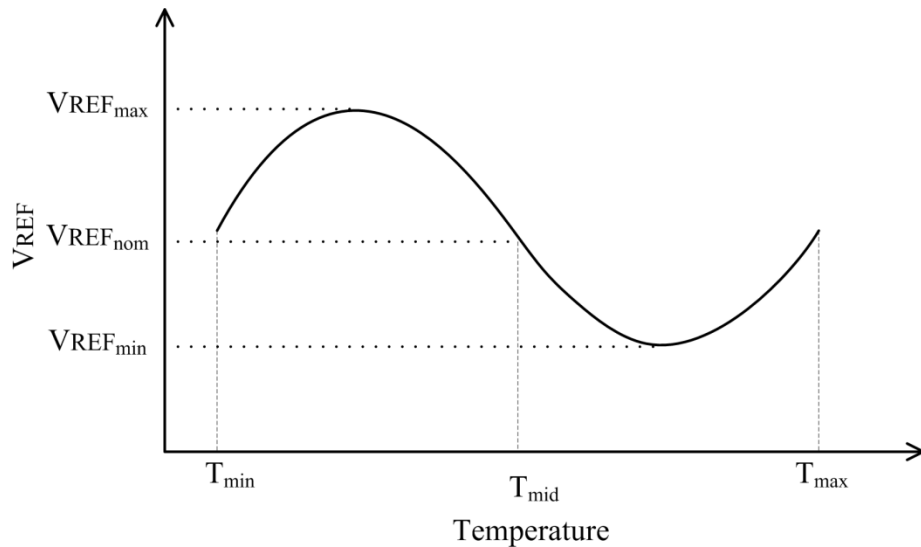


(2) PTAT (Proportional to absolute temperature)



Terminologies

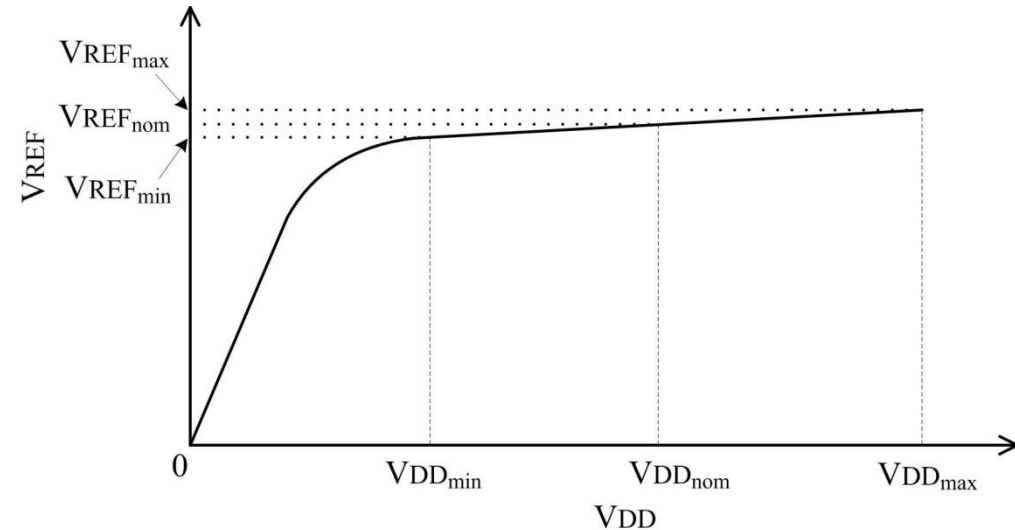
(3) Temperature coefficient (TC)



$$TC = \frac{V_{REF_{max}} - V_{REF_{min}}}{V_{REF_{nom}} (T_{max} - T_{min})} 10^6 \text{ (ppm/}^\circ\text{C)}$$

(Smaller the better)

(4) Line regulation (LR)

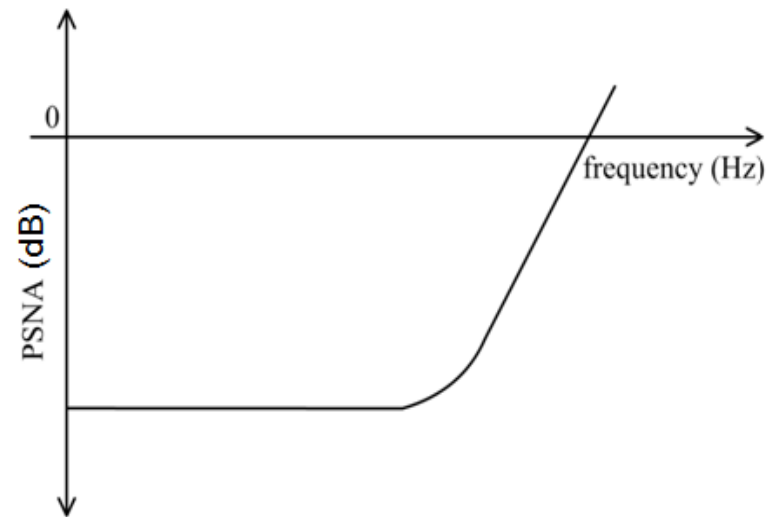


$$LR = \frac{V_{REF_{max}} - V_{REF_{min}}}{(V_{DD_{max}} - V_{DD_{min}})} \text{ (mV/V)}$$

(Smaller the better)

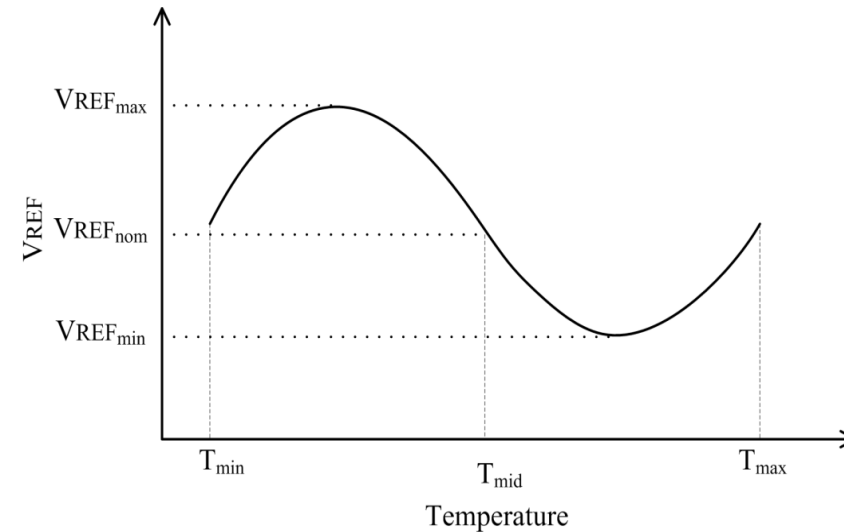
Terminologies

(5) Power supply noise attenuation (PSNA)



(Smaller the better)

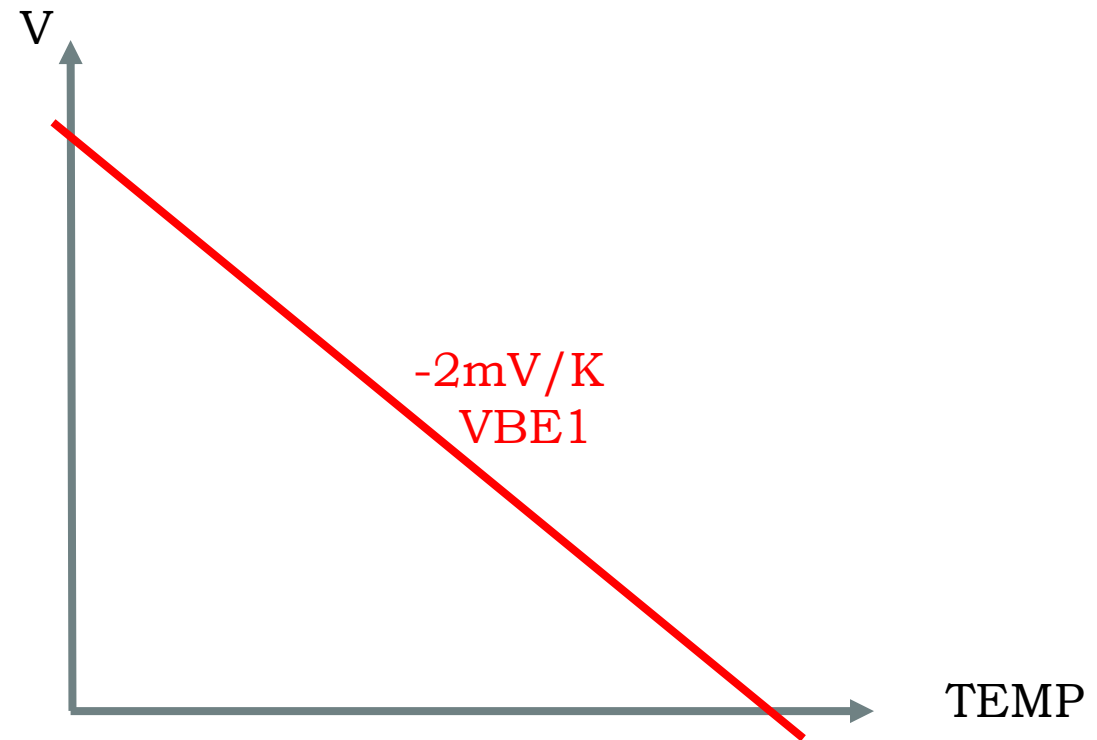
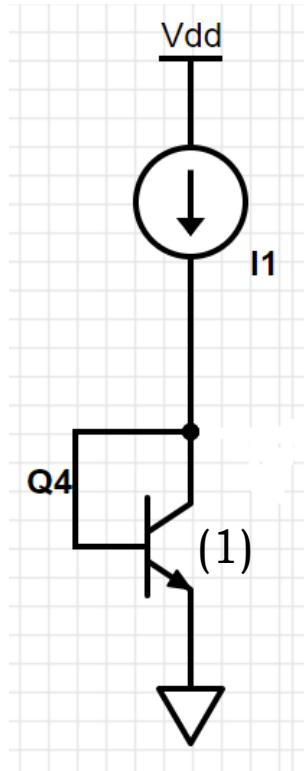
(6) Maximum absolute percentage variation (MAPV)



$$MAPV = \frac{\Delta V_{REF}}{V_{REF_{nom}}} \quad (\%)$$

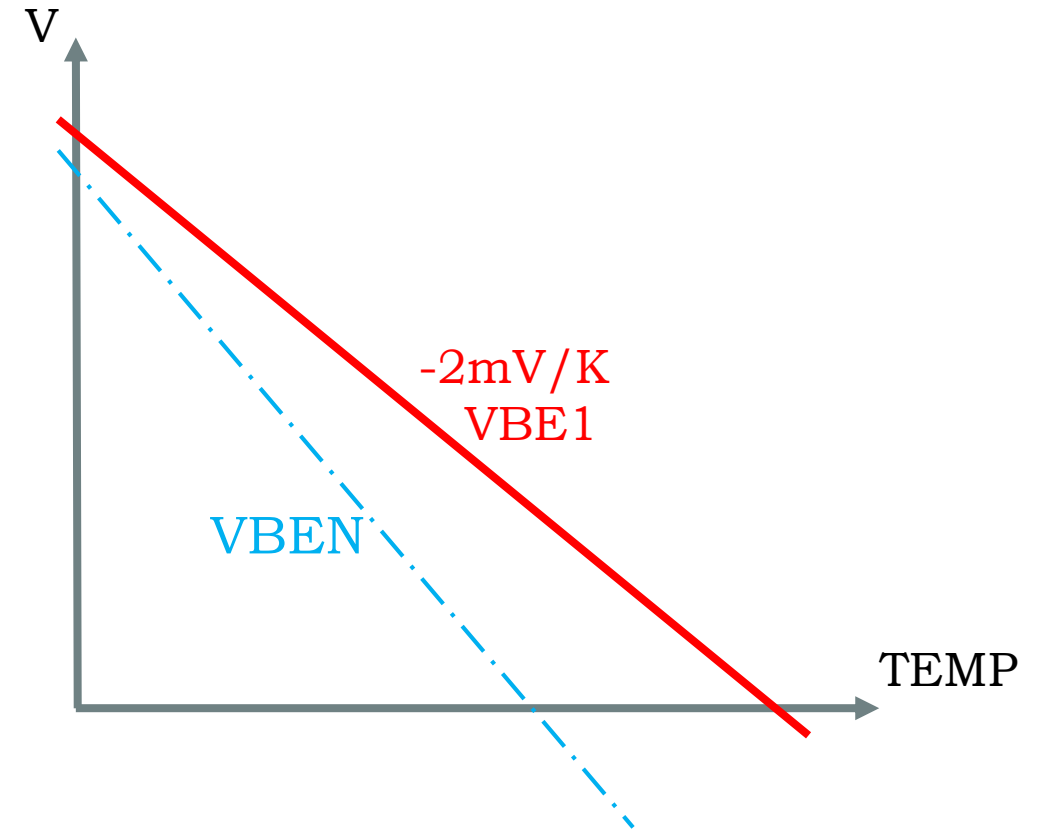
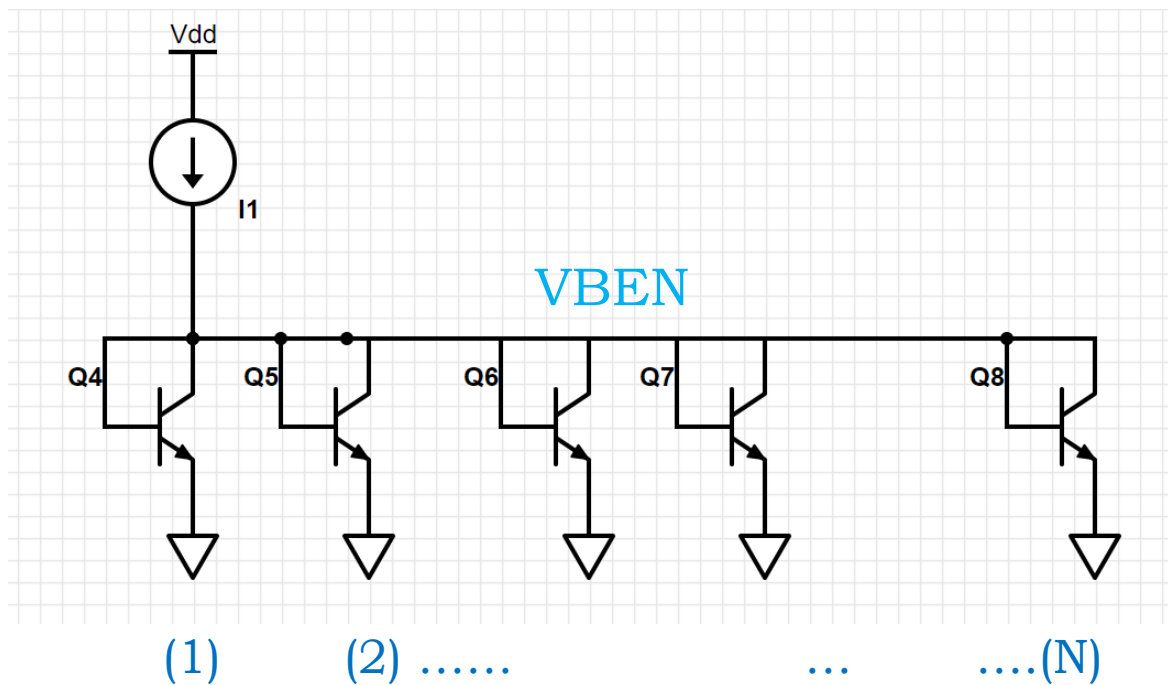
(Smaller the better)

Basic design concept



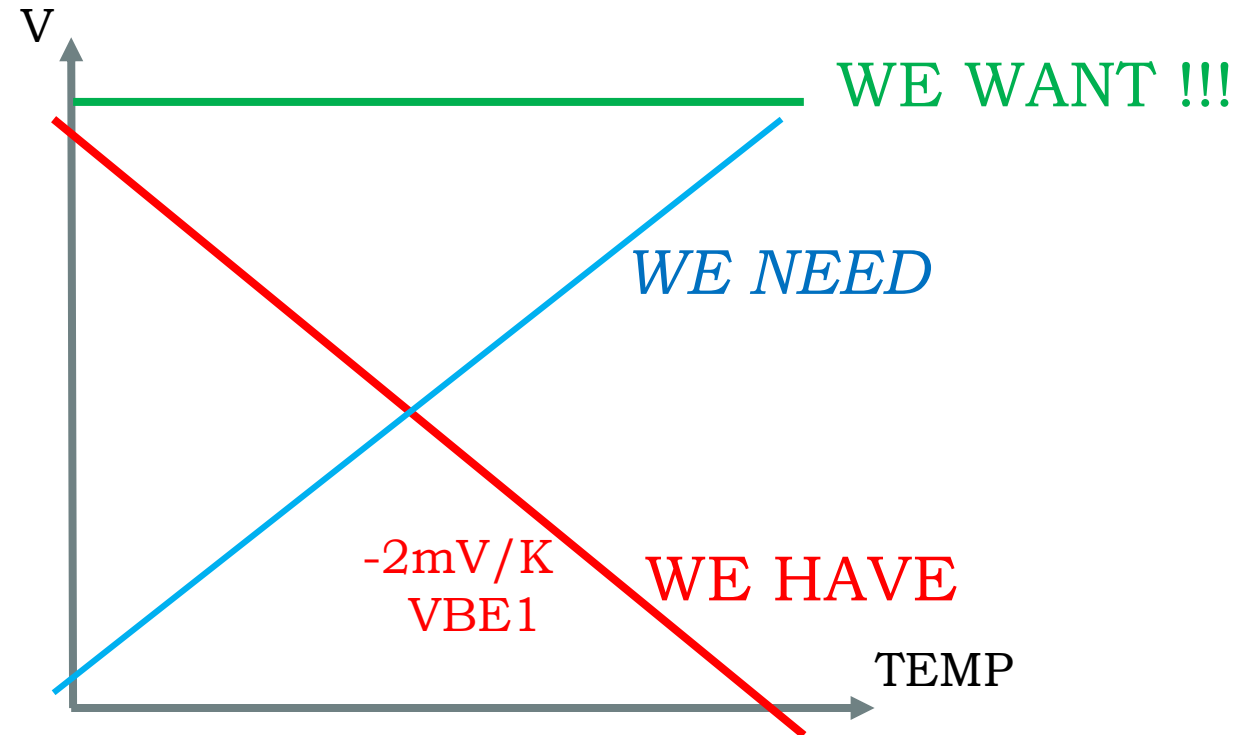
PNP : Parasitic BJT

Basic design concept



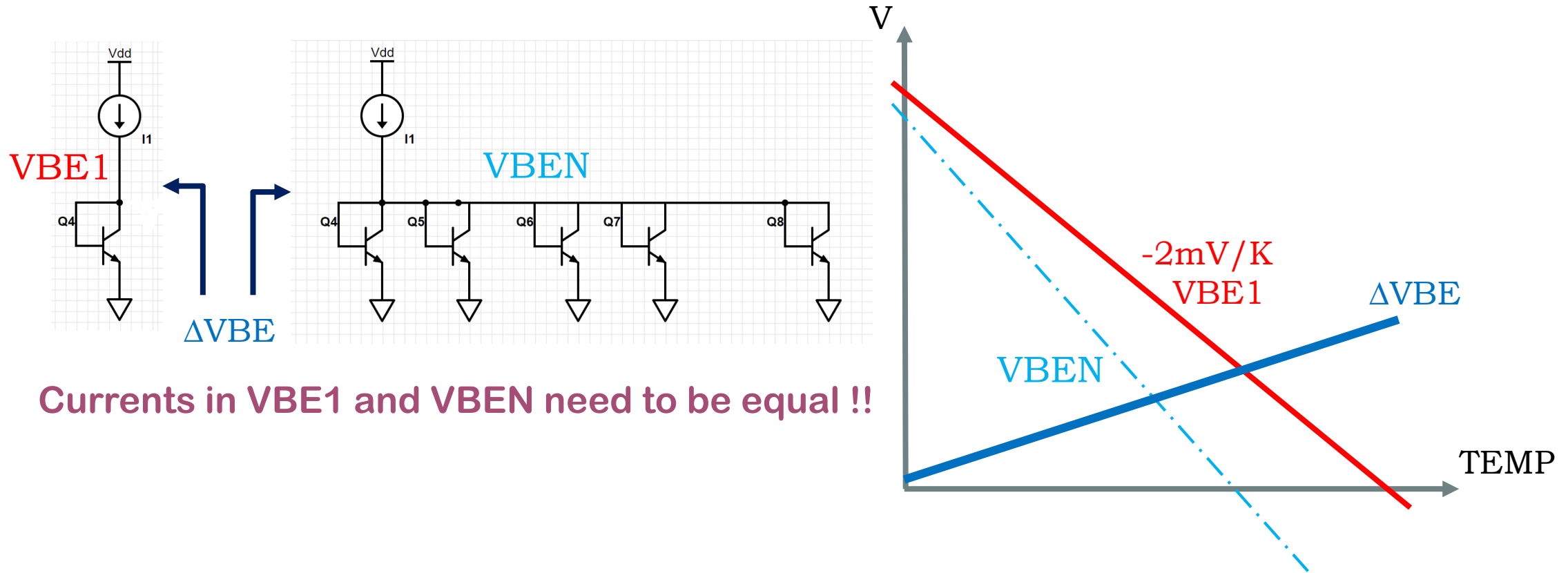
$VBEN$ is much steeper than $VBE1$ w.r.t. temperature

Basic design concept



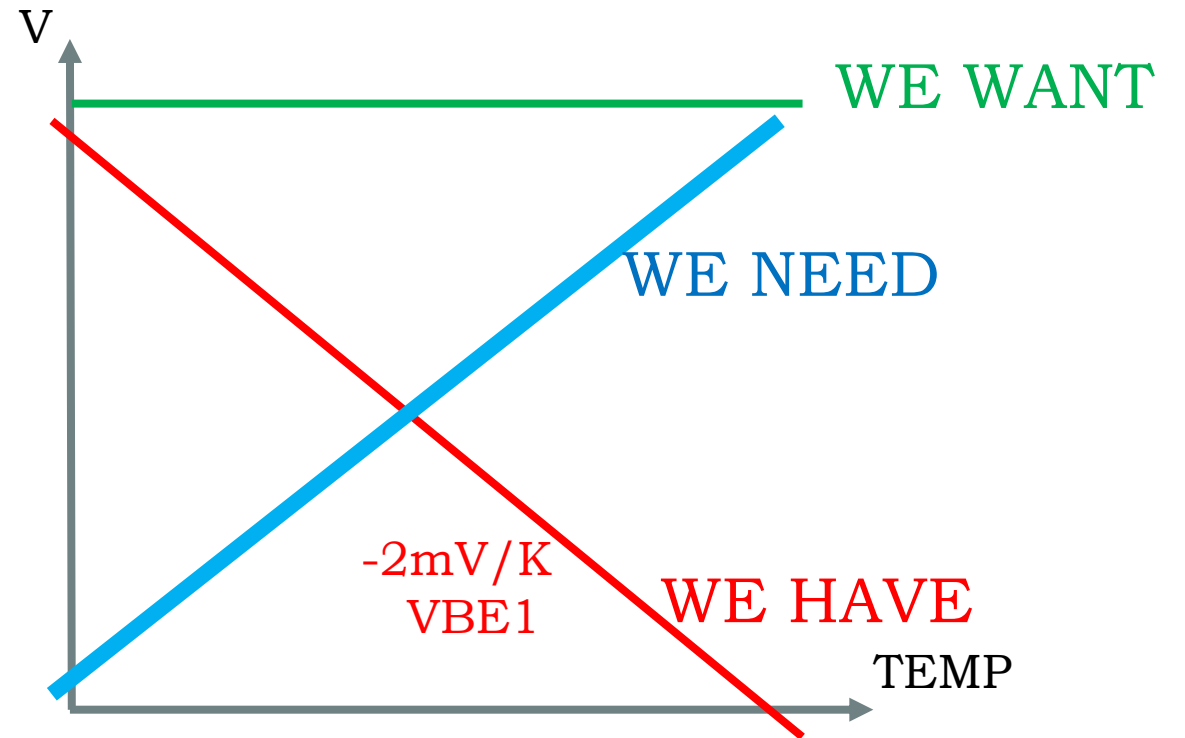
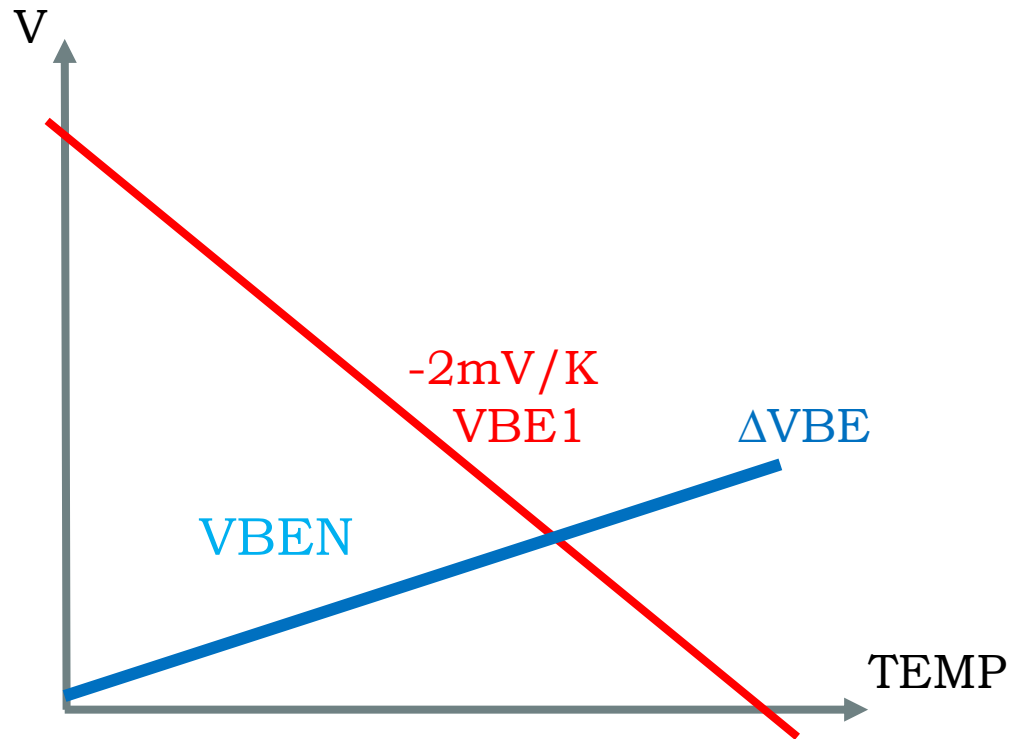
$$\text{WE HAVE} + \text{WE NEED} = \text{WE WANT !!!}$$

Basic design concept



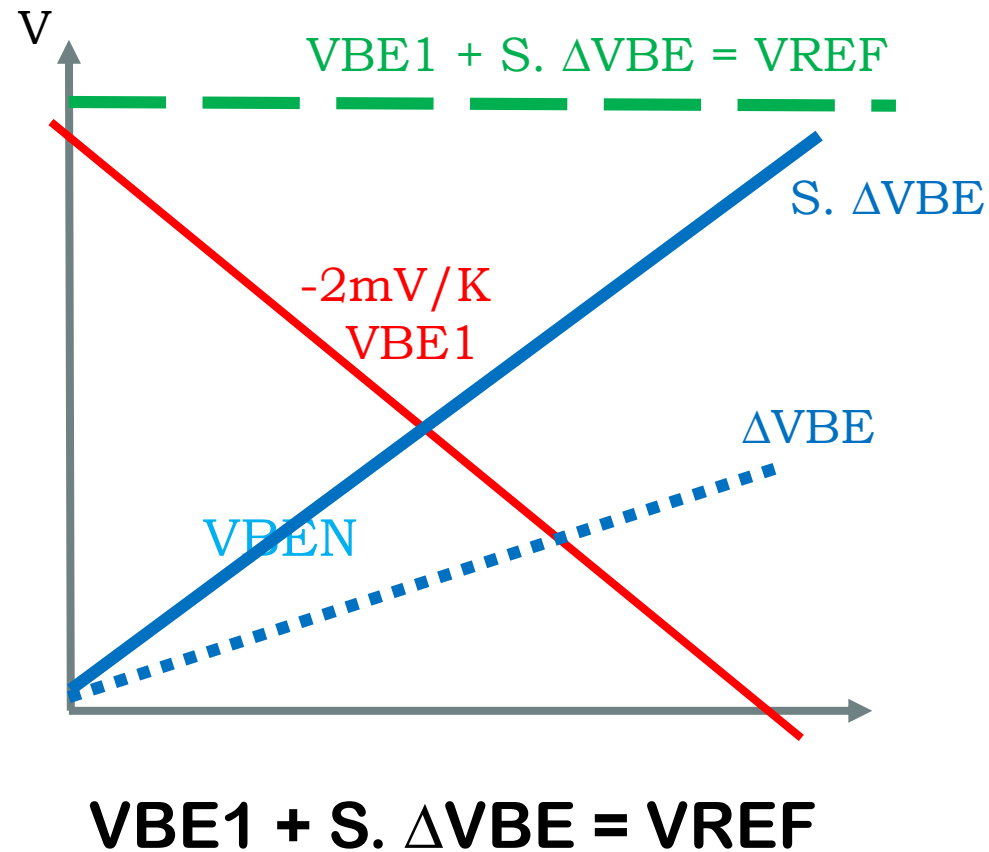
ΔV_{BE} has **positive slope** w.r.t. temperature

Basic design concept



ΔV_{BE} need to be scaled

Basic design concept



Basic design concept

$$V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$$

How to find value of S ????

Basic design concept

$$V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$$

$$d/dT (V_{BE1} + S \cdot \Delta V_{BE}) = d/dT (V_{REF})$$

$$d/dT (V_{BE1}) = -2 \text{ mV/K}$$

$$S \cdot d/dT (\Delta V_{BE}) = 2 \text{ mV/K}$$

$$S \cdot d/dT (V_T \ln N) = 2 \text{ mV/K}$$

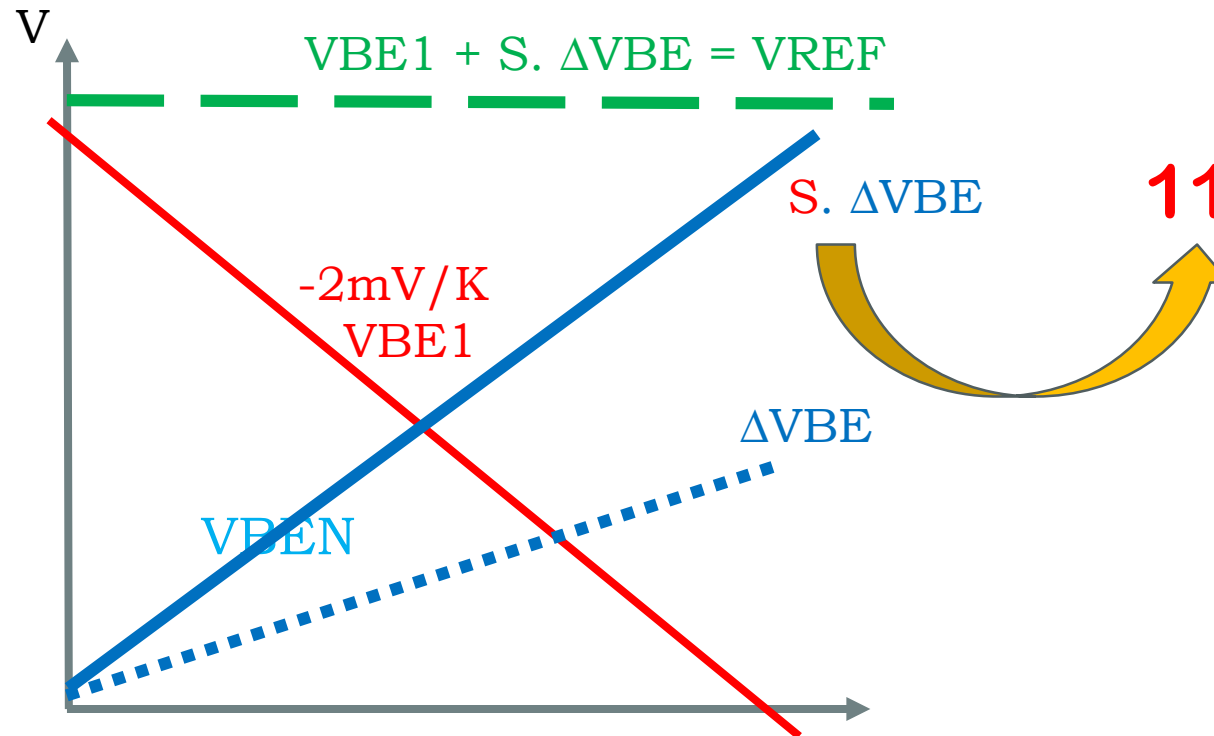
If $N=8$ and T at Room temperature

$$S = 11$$

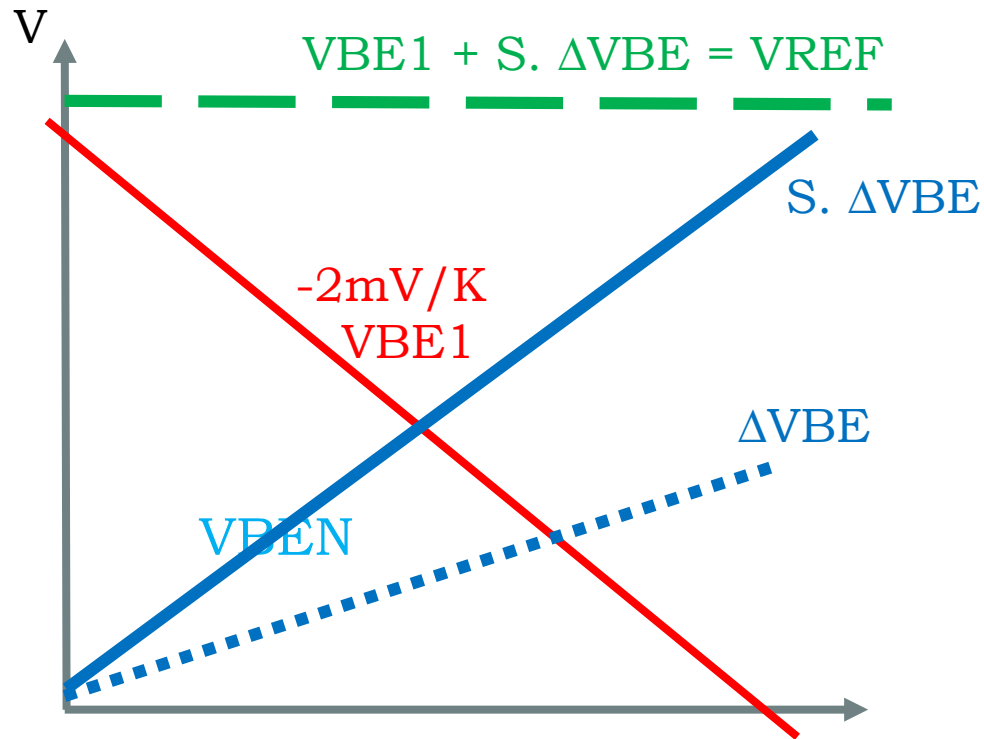
Therefore, $V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$

$$700\text{mV} + 11 \cdot (26\text{mV} \ln 11) = 1300 \text{ mV} \text{ Bandgap !!!!}$$

Basic design concept



Basic design concept



$$V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$$

$$V_{BE1} = V_A$$

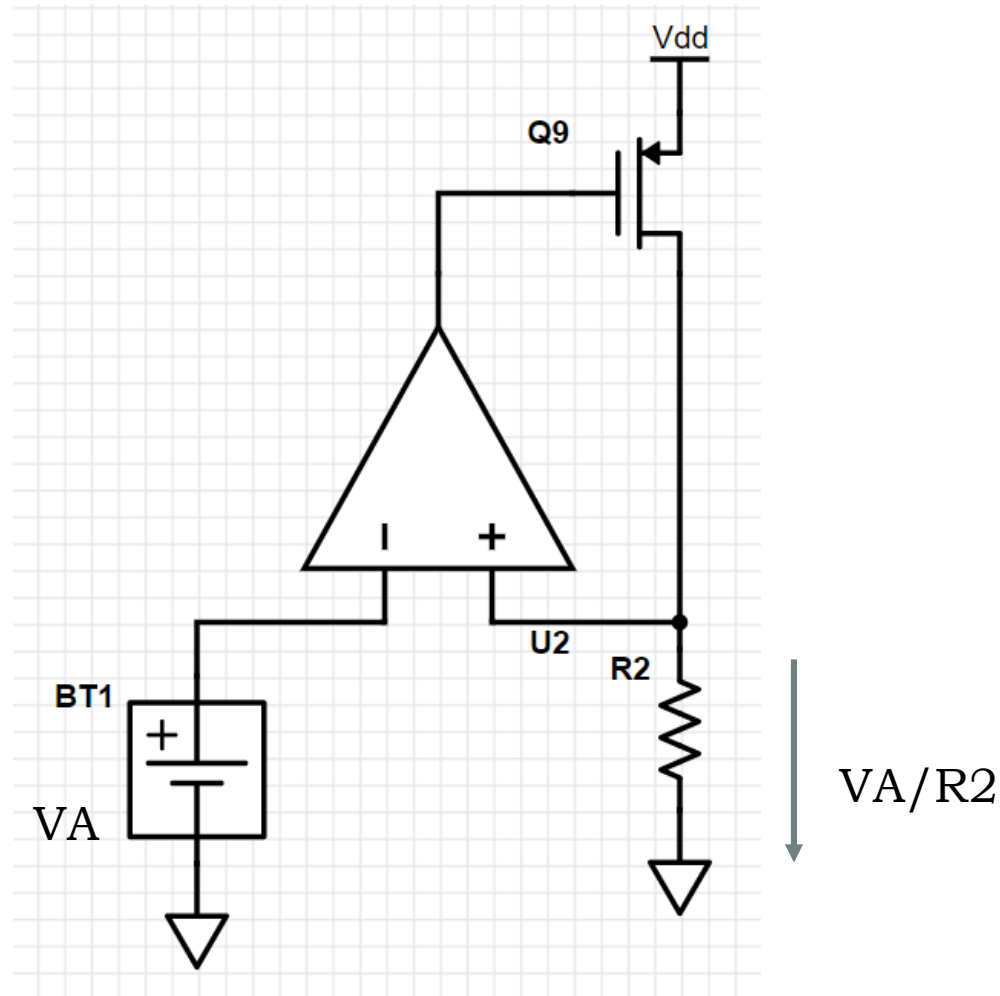
$$V_{BEN} = V_B$$

$$V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$$

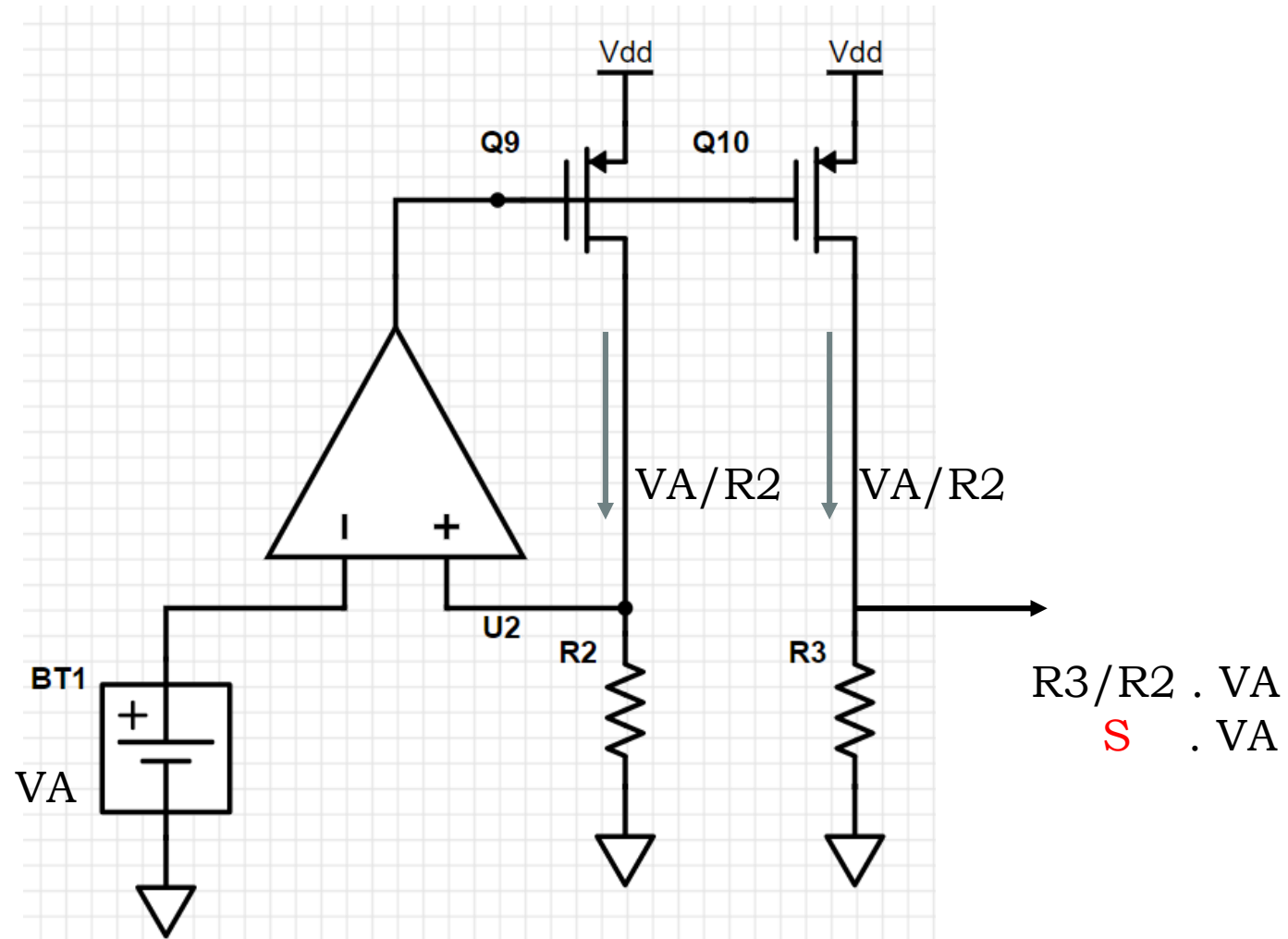
$$V_A + S \cdot (V_A - V_B) = V_{REF}$$

$$V_A + S \cdot V_A - S \cdot V_B = V_{REF}$$

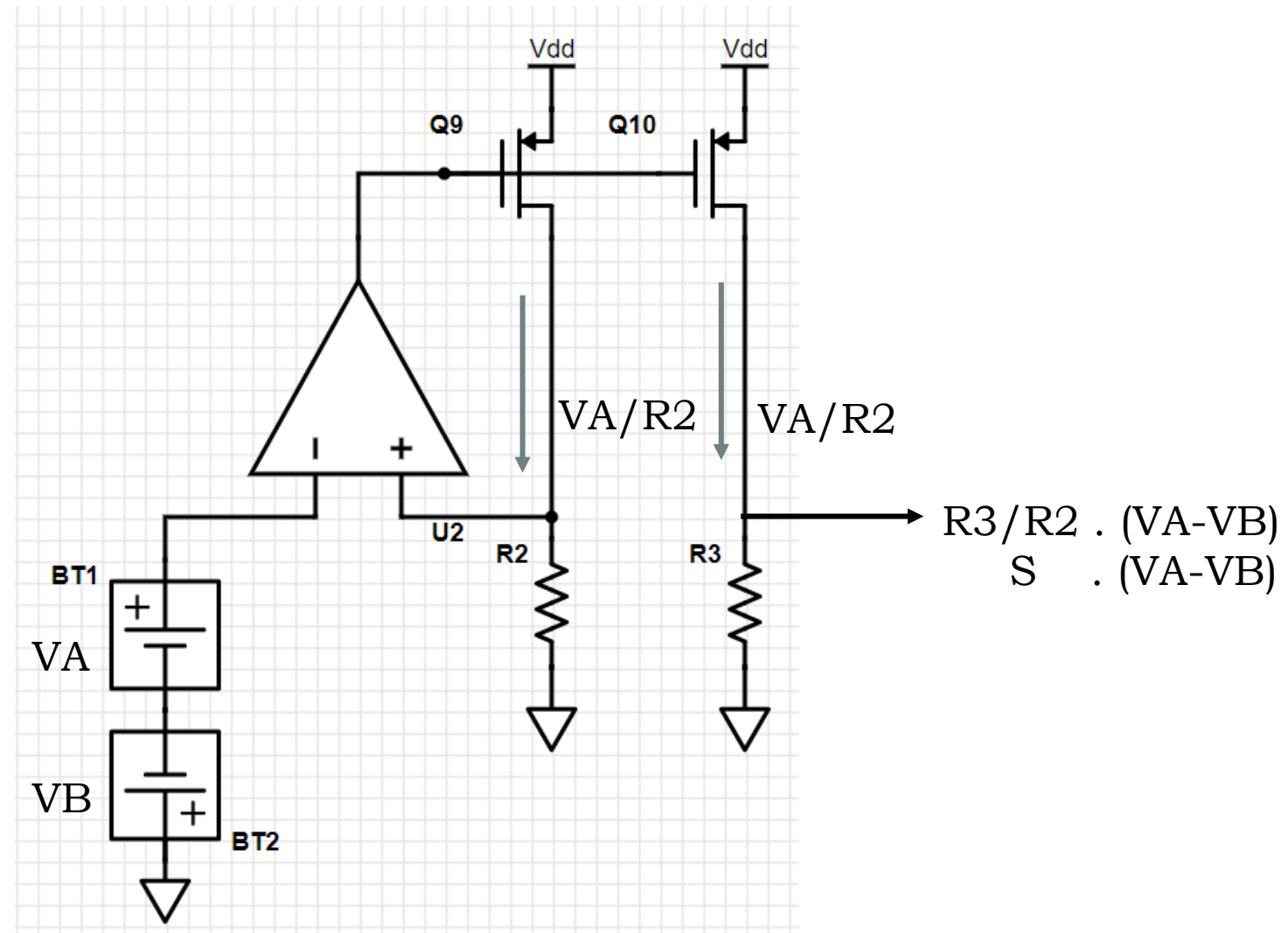
Basic design concept realization



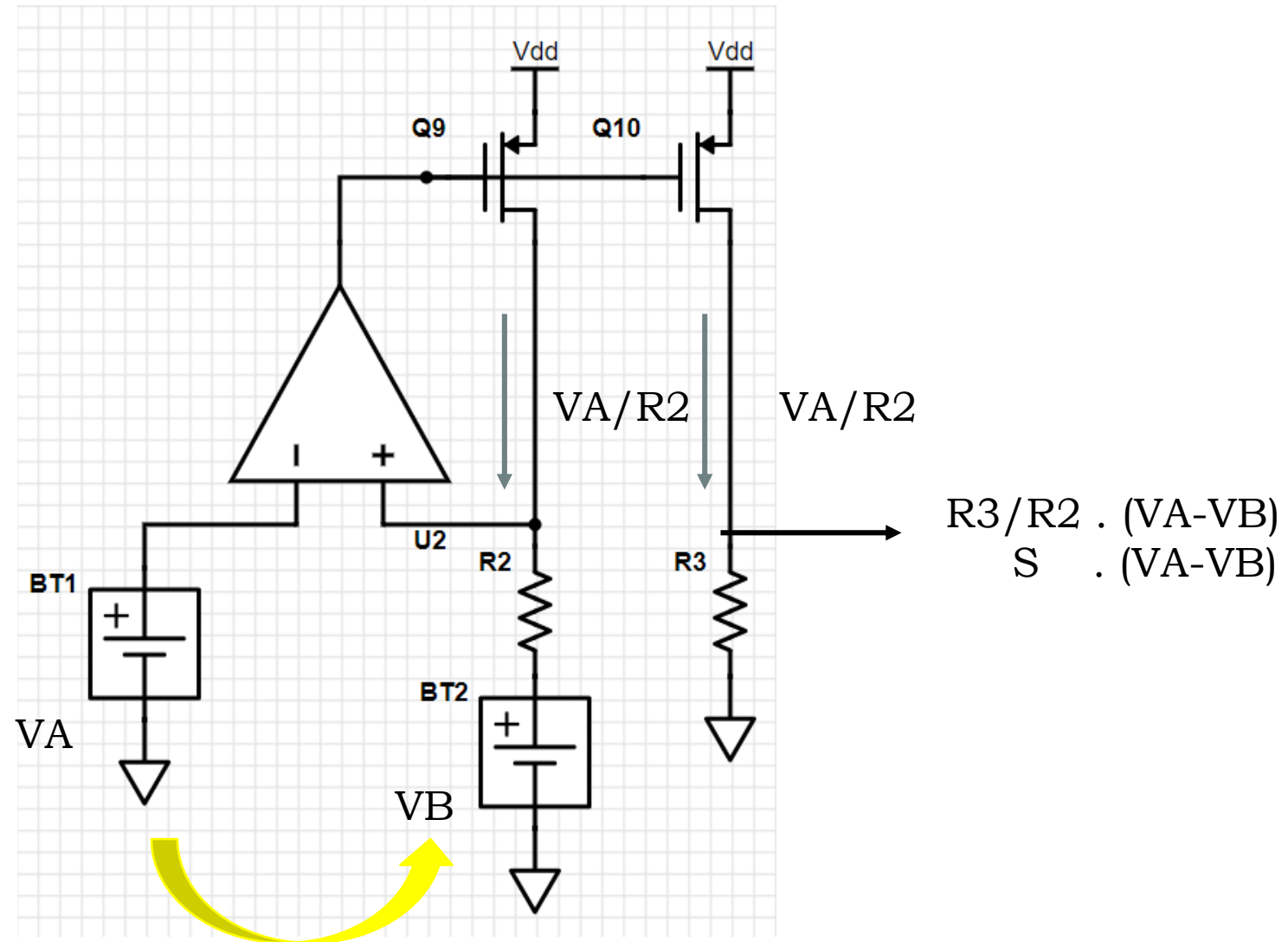
Basic design concept realization



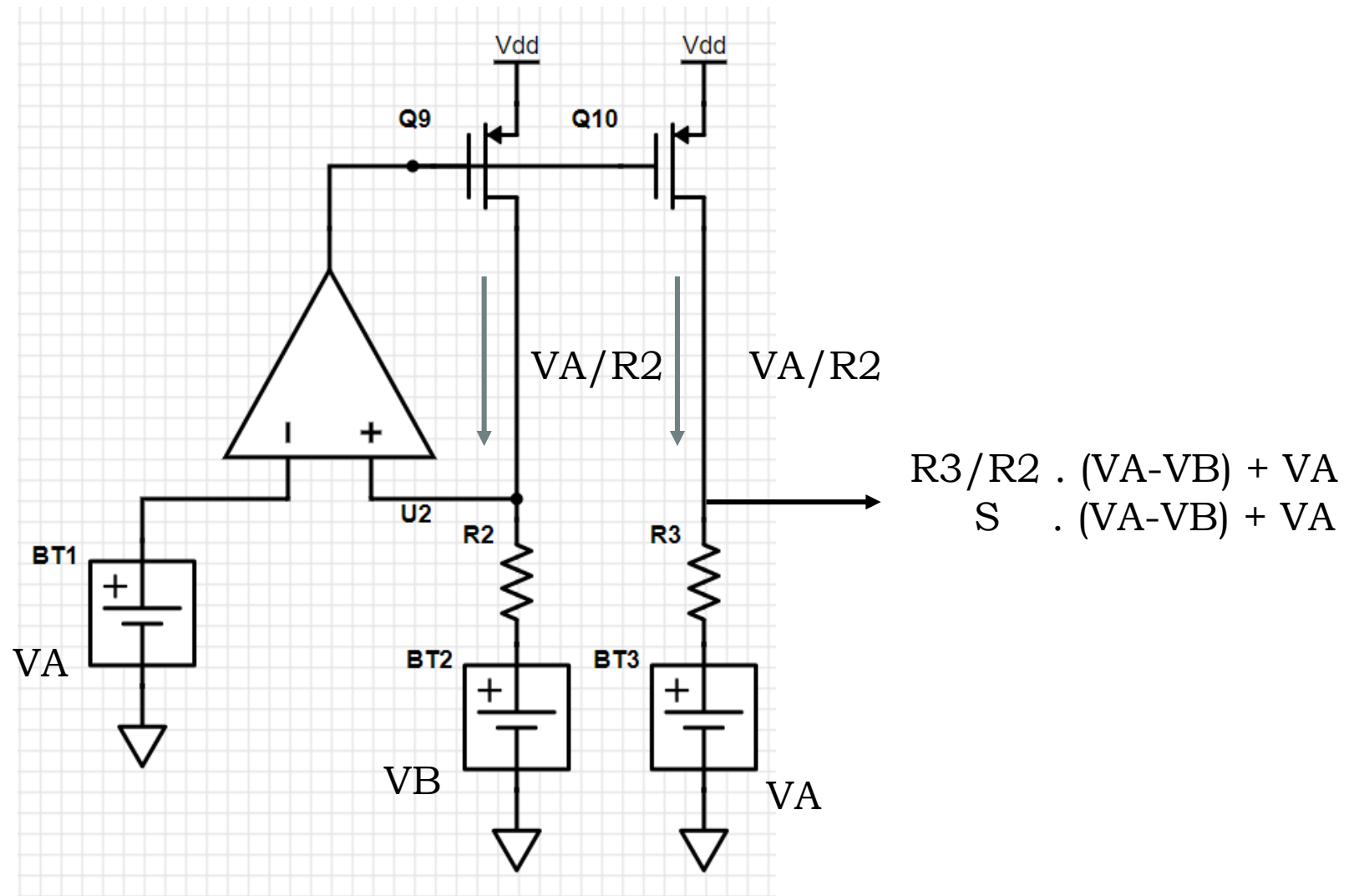
Basic design concept realization



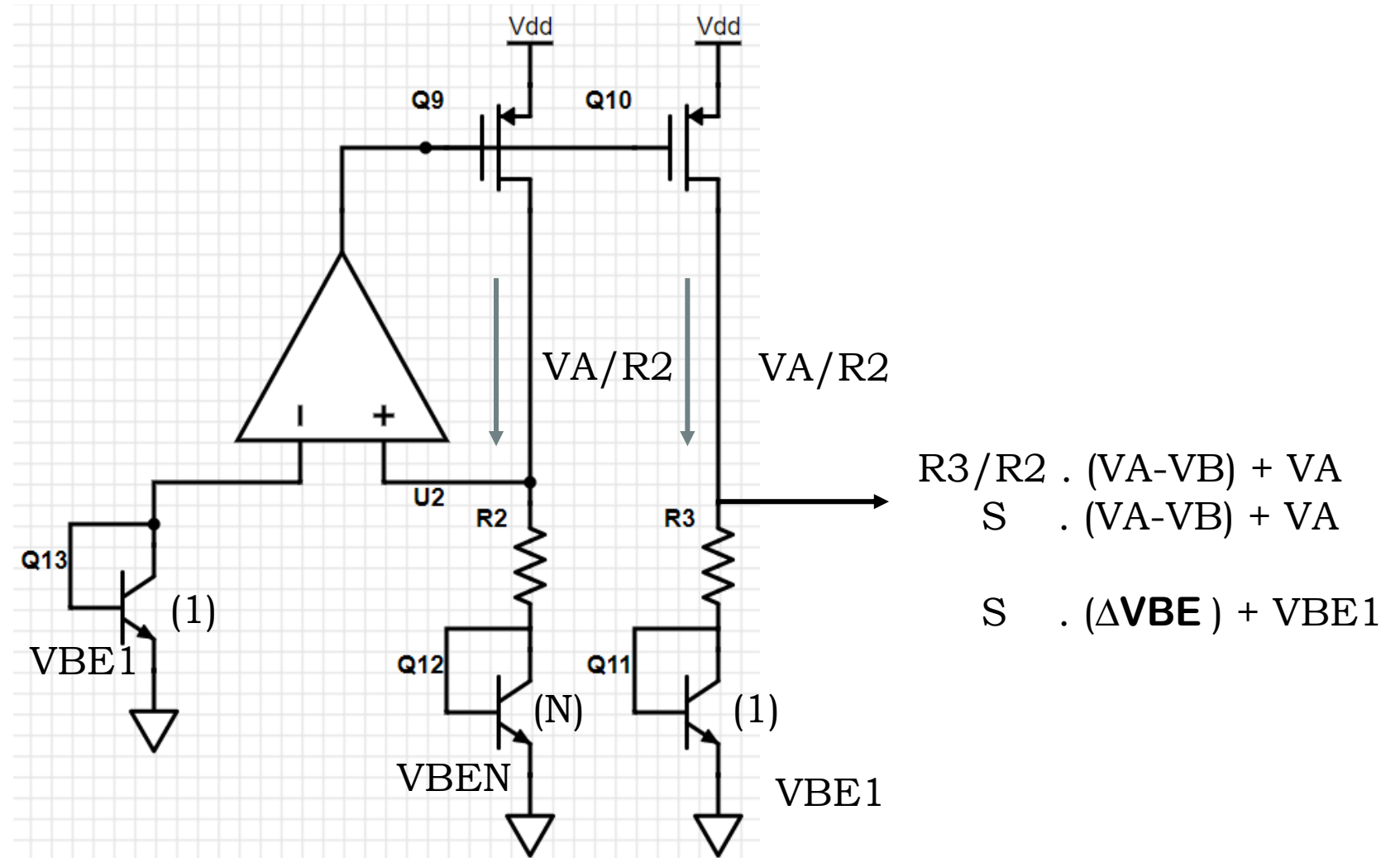
Basic design concept realization



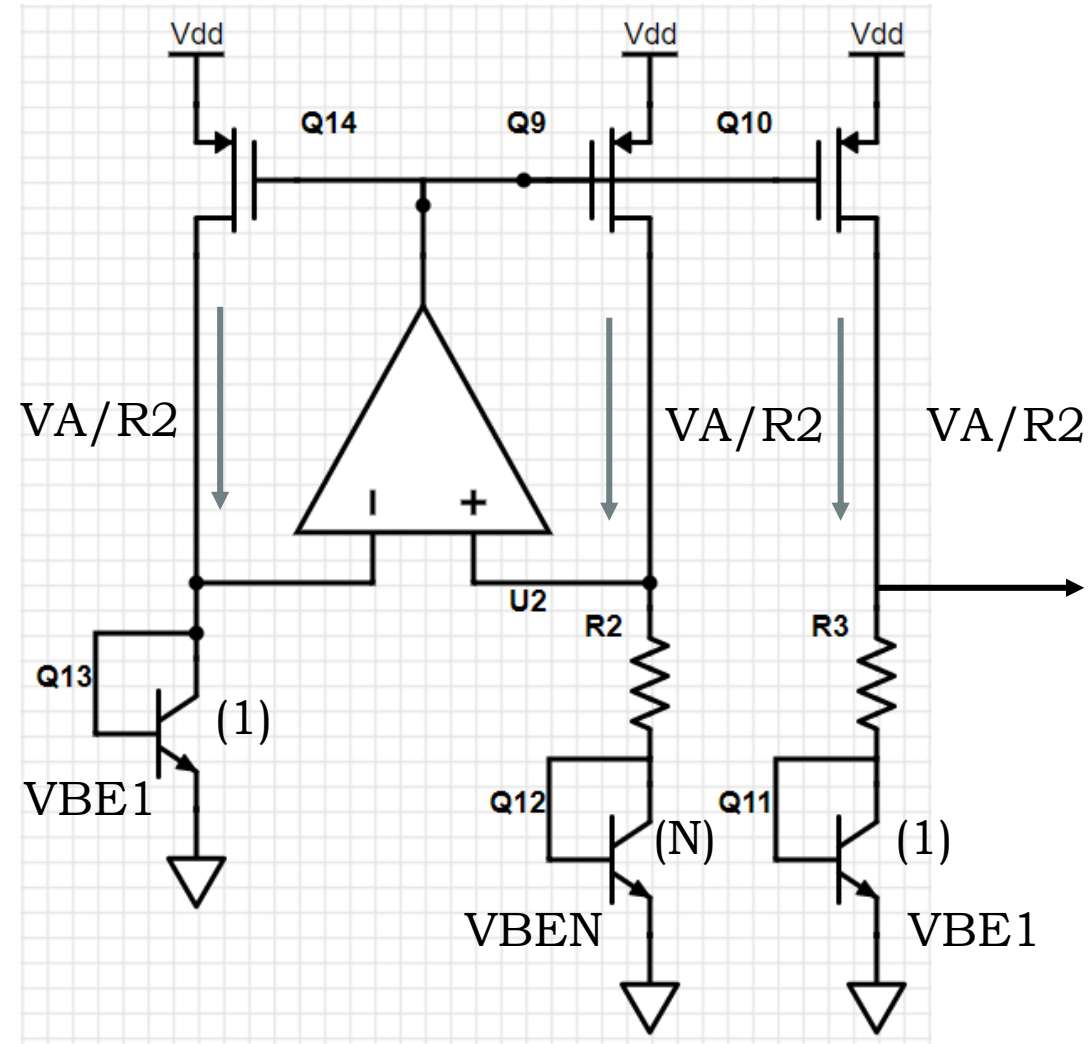
Basic design concept realization



Basic design concept realization



Basic design concept realization



$$\begin{aligned} & \frac{R3}{R2} \cdot (V_A - V_B) + V_A \\ & S \cdot (V_A - V_B) + V_A \\ & S \cdot (\Delta V_{BE}) + V_{BE1} \end{aligned}$$

Another Approach !!

Basic design concept realization

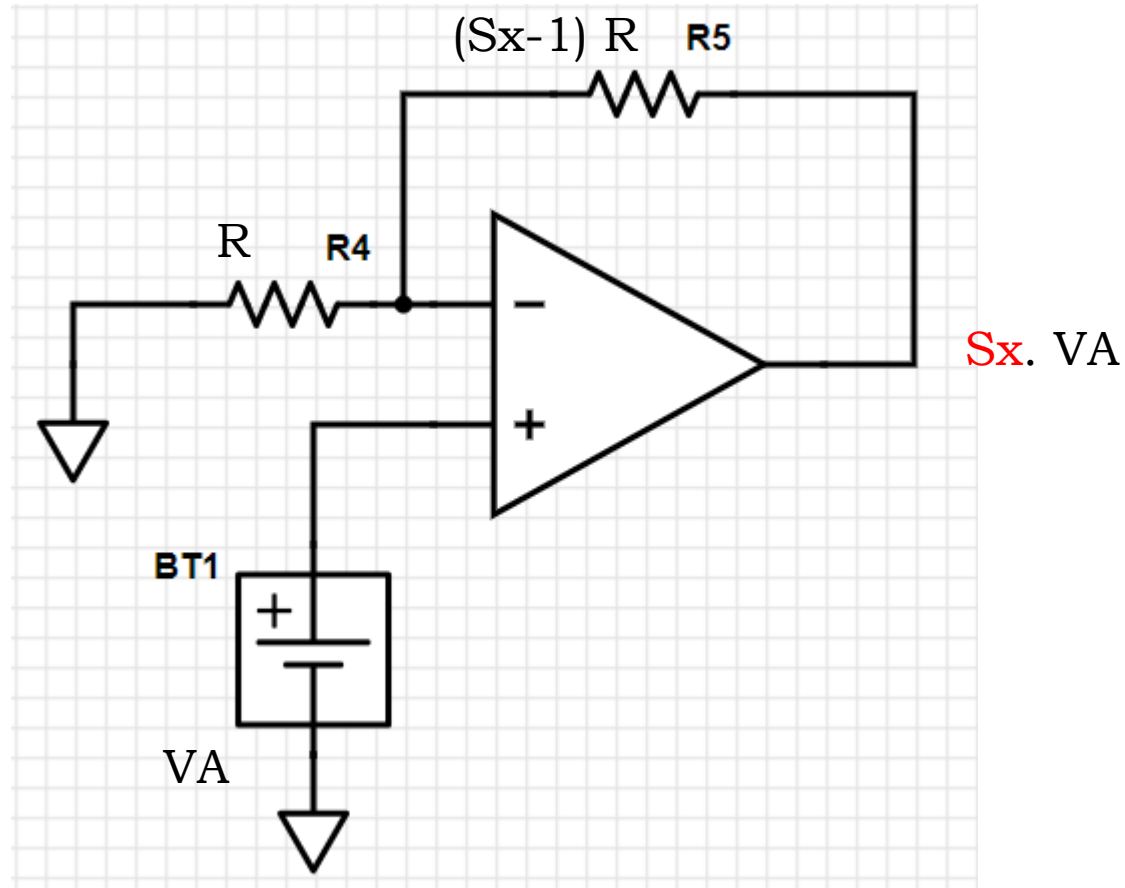
$$V_{BE1} + S \cdot \Delta V_{BE} = V_{REF}$$

$$V_{BE1} = V_A$$

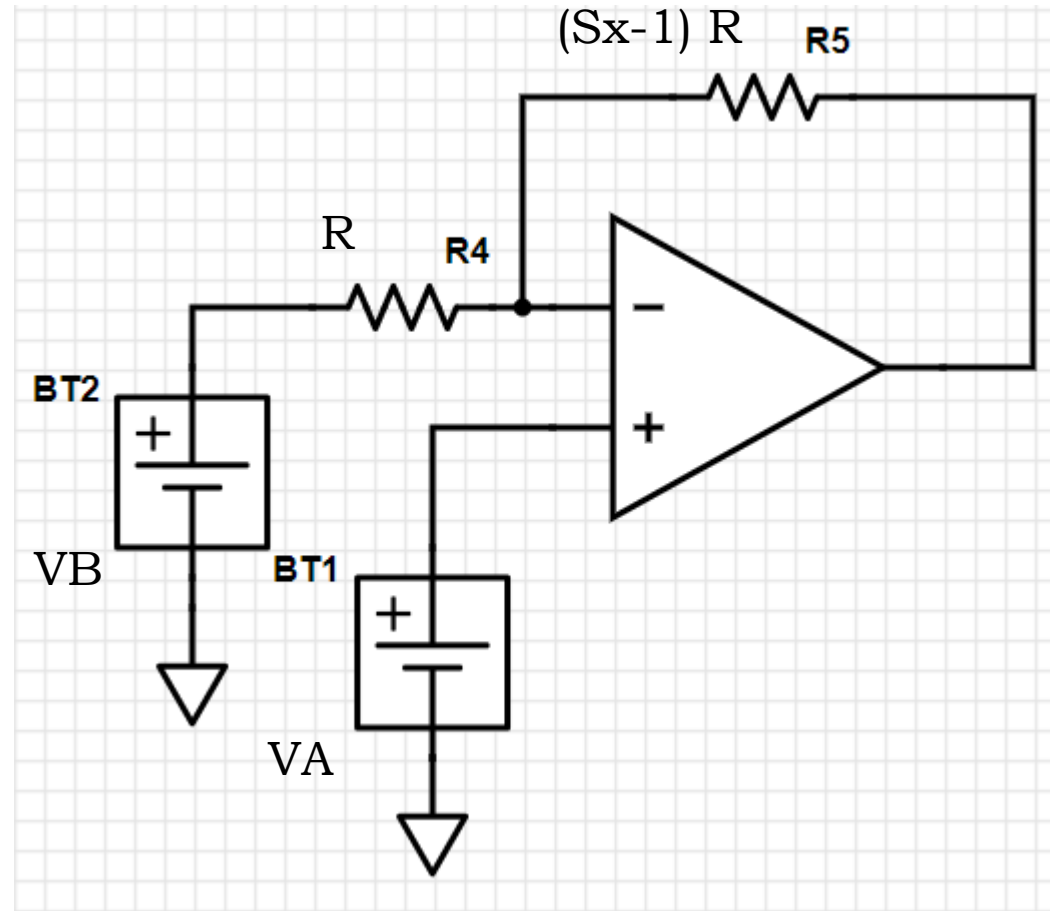
$$V_{BEN} = V_B$$

$$V_A + S \cdot (V_A - V_B)$$

$$V_A + S \cdot V_A - S \cdot V_B$$

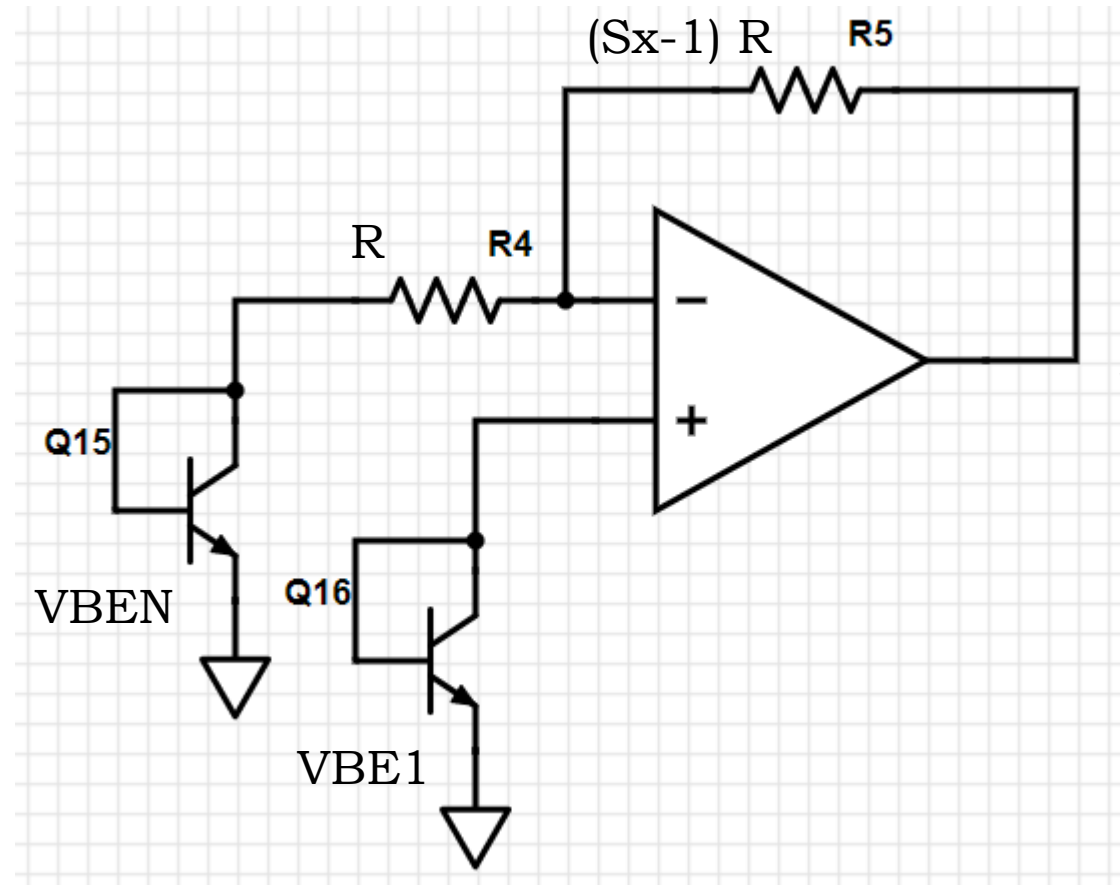


Basic design concept realization



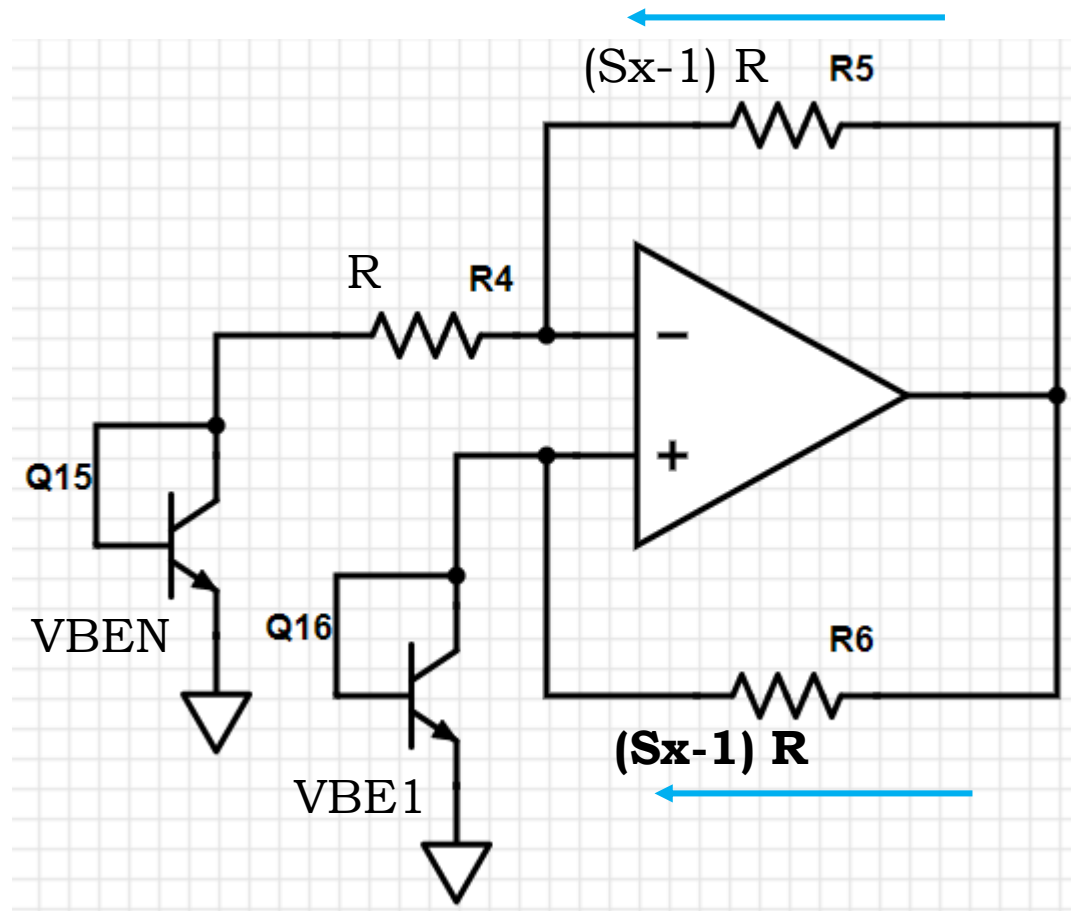
$$Sx \cdot V_A + (-(Sx-1) \cdot V_B)$$
$$V_A + (Sx-1) \cdot (V_A - V_B)$$

Basic design concept realization



$$\begin{aligned} &S_x \cdot V_{BE1} + -(S_x - 1) \cdot V_{BEN} \\ &V_{BE1} + (S_x - 1) \cdot (V_{BE1} - V_{BEN}) \end{aligned}$$

Basic design concept realization



$$\begin{aligned} &Sx \cdot VBE1 + -(Sx-1) \cdot VBEN \\ &VBE1 + (Sx-1) \cdot (VBE1 - VBEN) \end{aligned}$$

Basic design concept realization

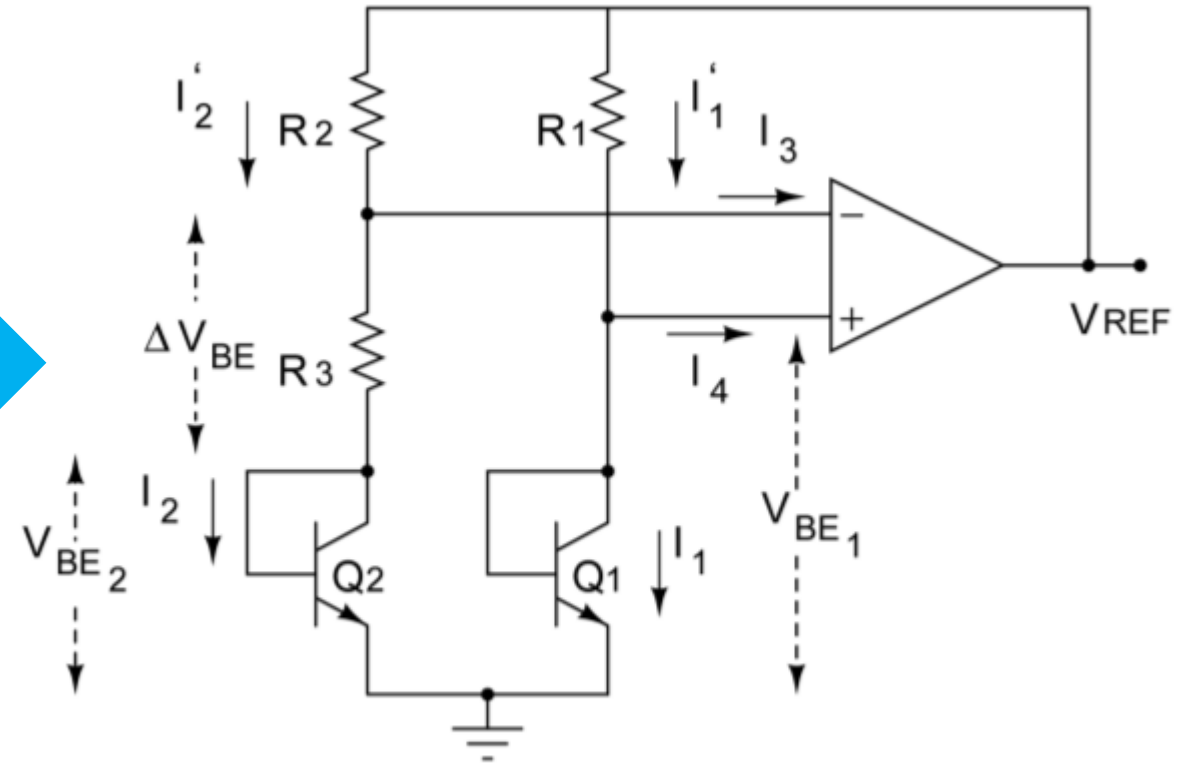
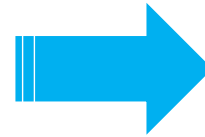
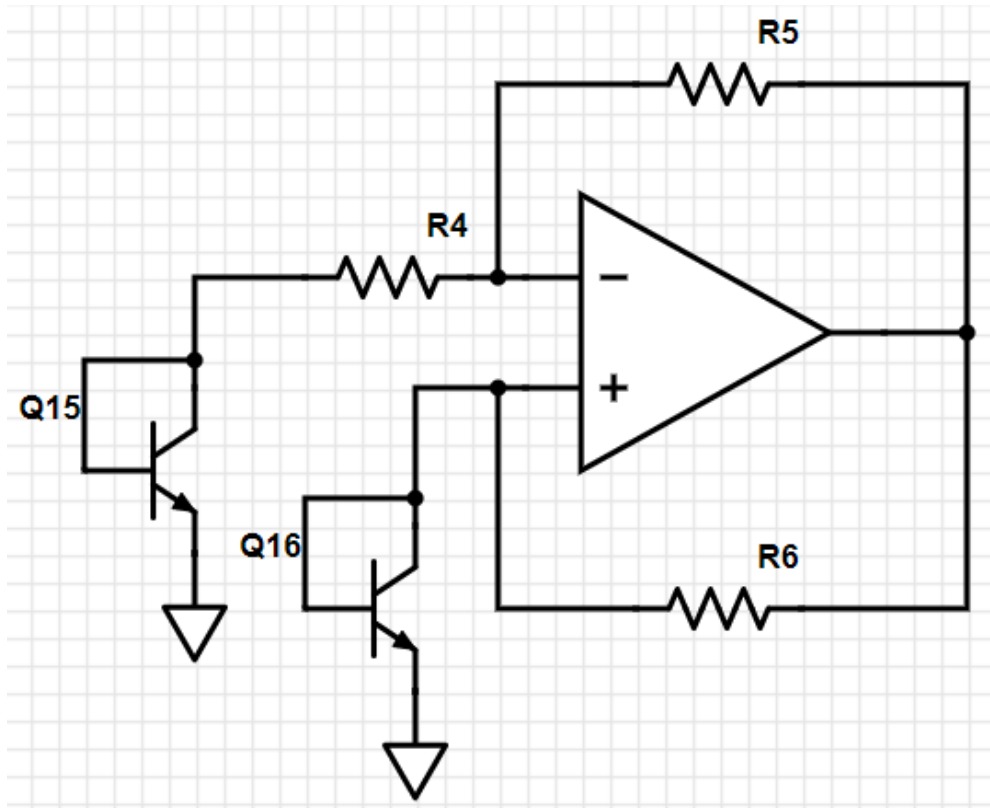


Figure . Kuijk voltage reference: 1973.

Requirement !!

Requirement

$$V_{BE1} \approx V_T \ln (I/I_s)$$

$$V_{BE1} = V_T \ln [(I/I_s) + 1]$$

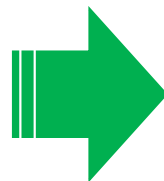
$$V_{BE2} = V_T \ln [(I/N.I_s) + 1]$$

$$\Delta V_{BE} = V_T \ln [N.(I+I_s)/(I+N.I_s)]$$

$$I = \Delta V_{BE}/R$$

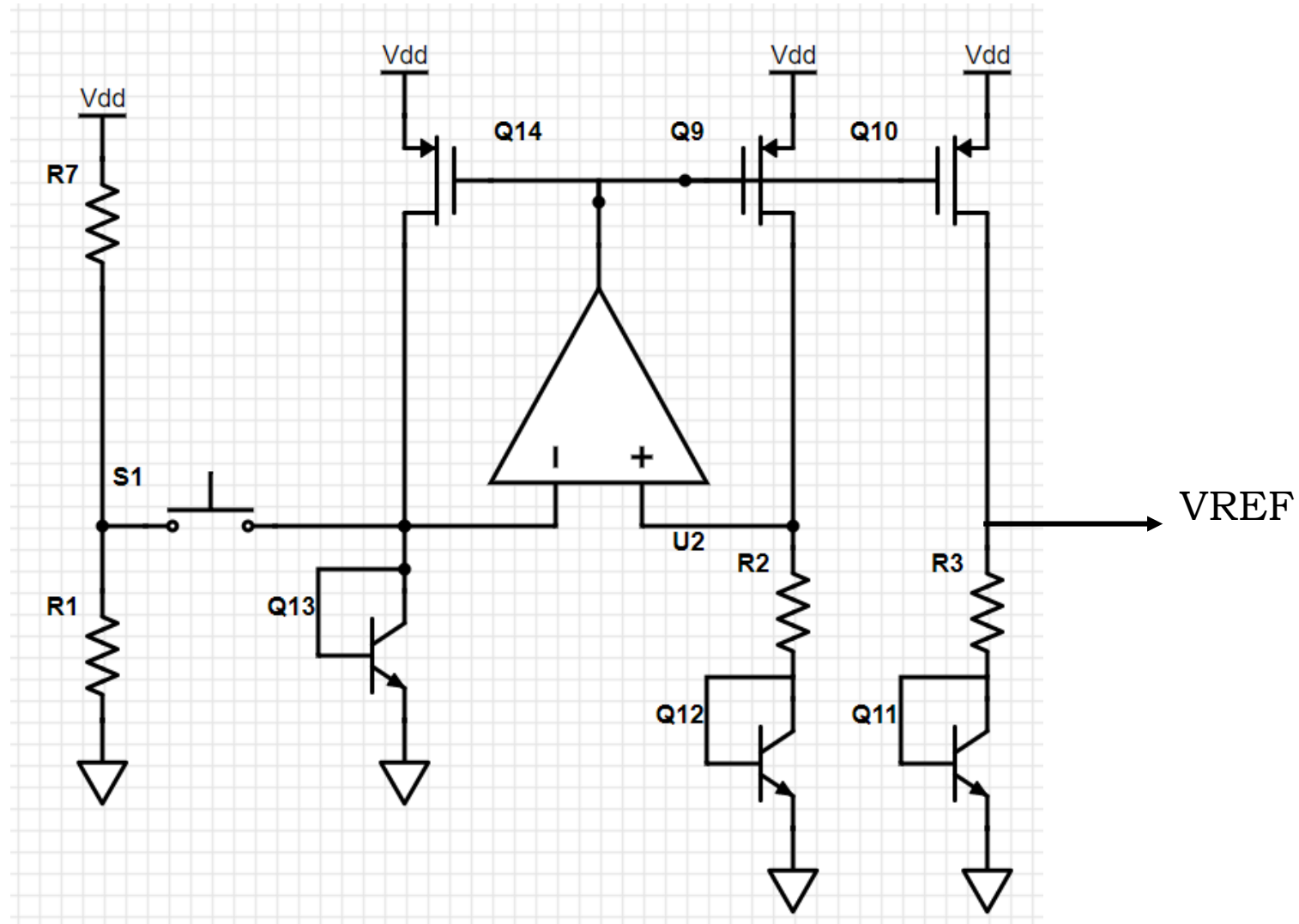
$$= V_T/R \ln [N.(I+I_s)/(I+N.I_s)]$$

2 Solutions !!!!

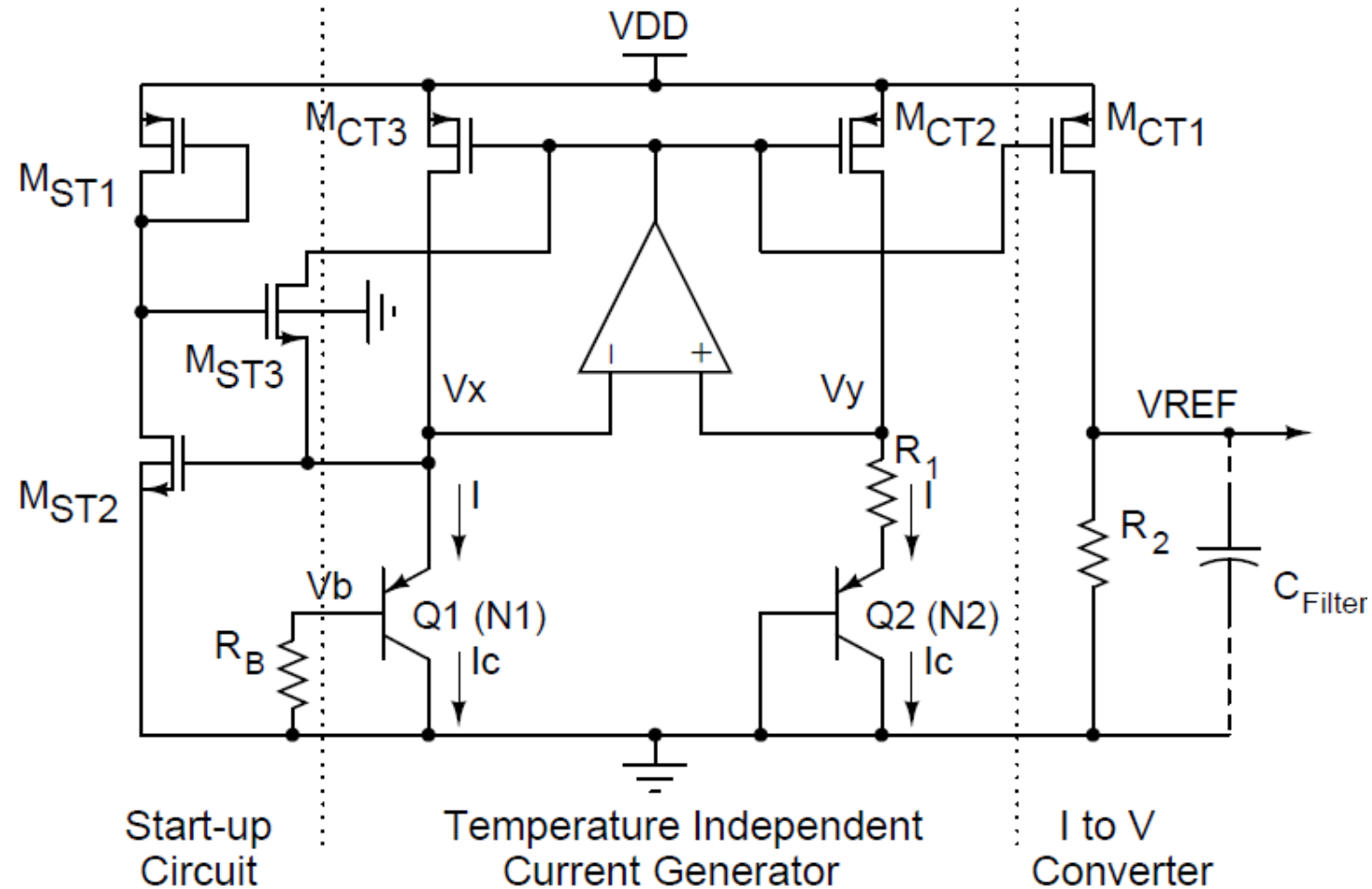


$I = 0$ and $I = \Delta V_{BE}/R$

Start-up requirement



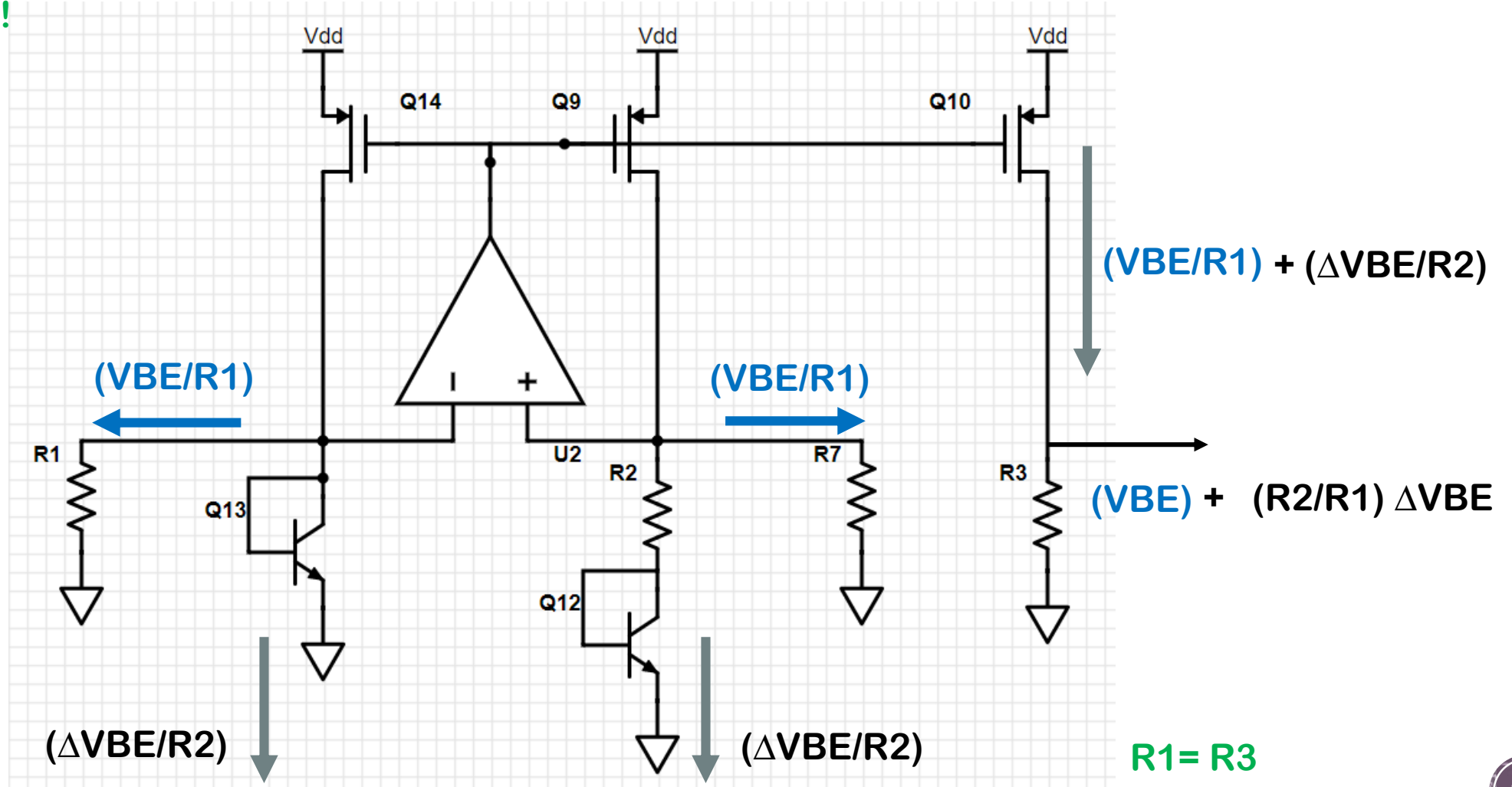
Example of Start-up



Popular Sub-1V BGR !!!

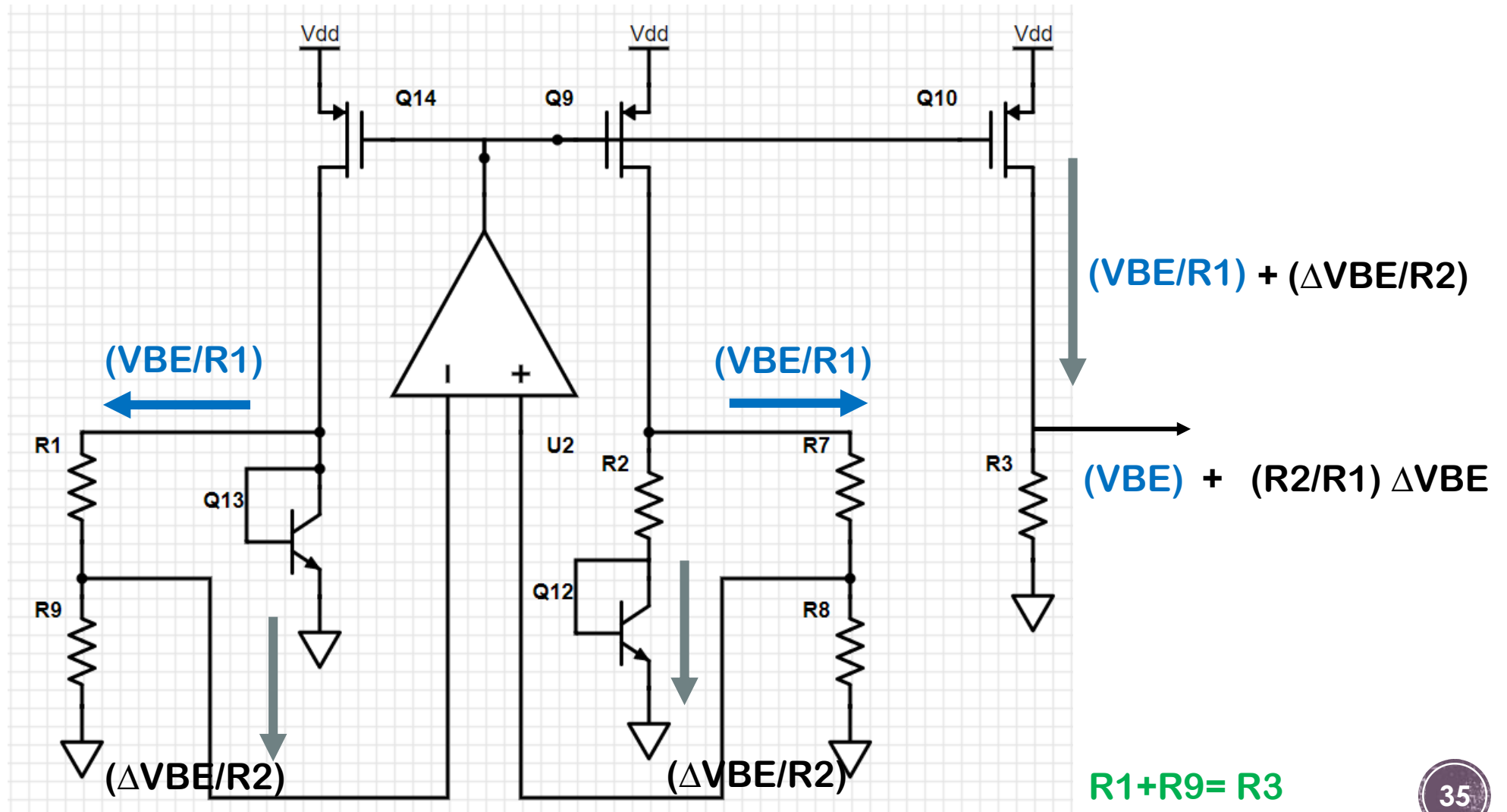
Popular Sub-1V BGR : Banba et al. JSSC'99

Current Mode !!!



Popular Sub-1V BGR : Modified, JSSC'02

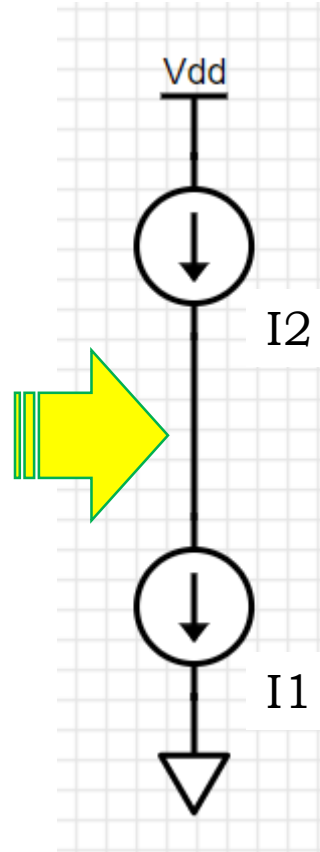
Current Mode !!!



One MORE Approach for BGR !!

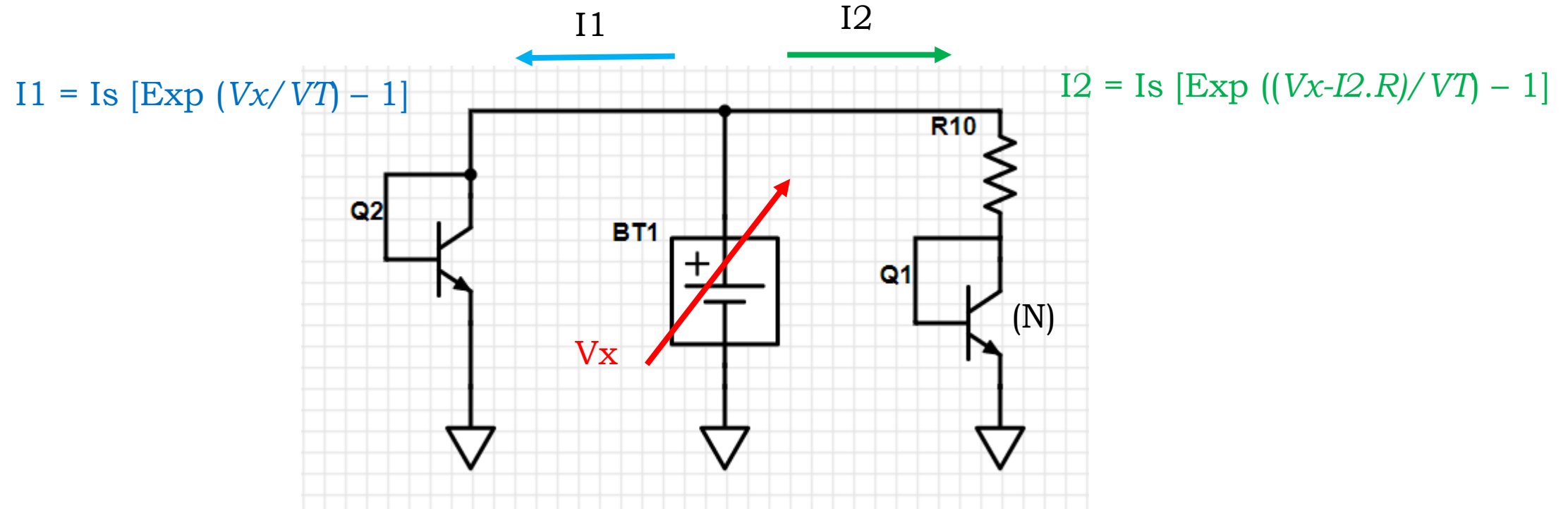
Basic design concept realization

Middle Potential ???



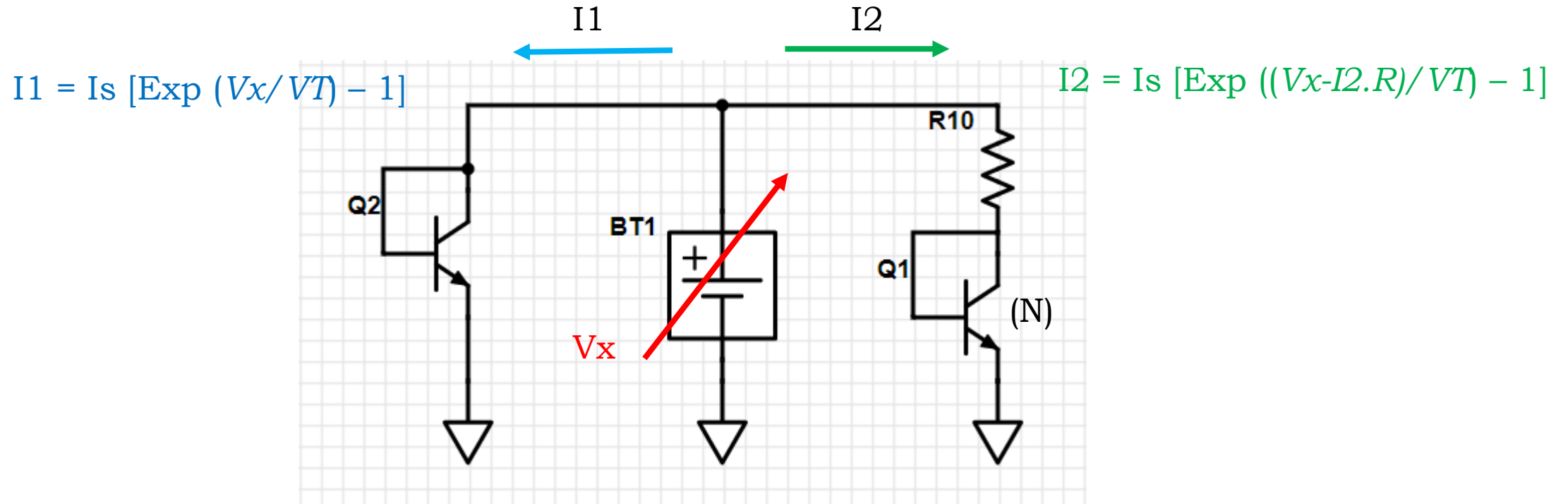
Basic design concept realization

$$I = I_s [\exp (V_{BE}/V_T) - 1]$$



Basic design concept realization

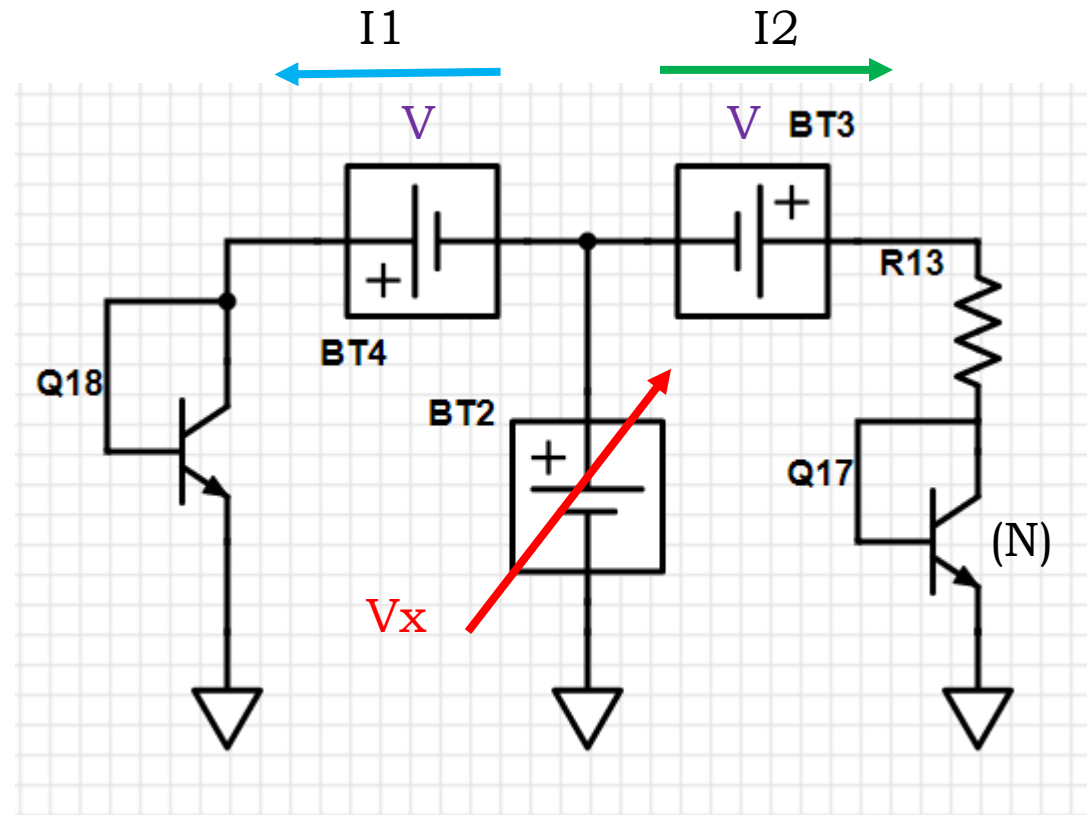
We have to make $I_1 = I_2 = I = \Delta V_{BE}/R$



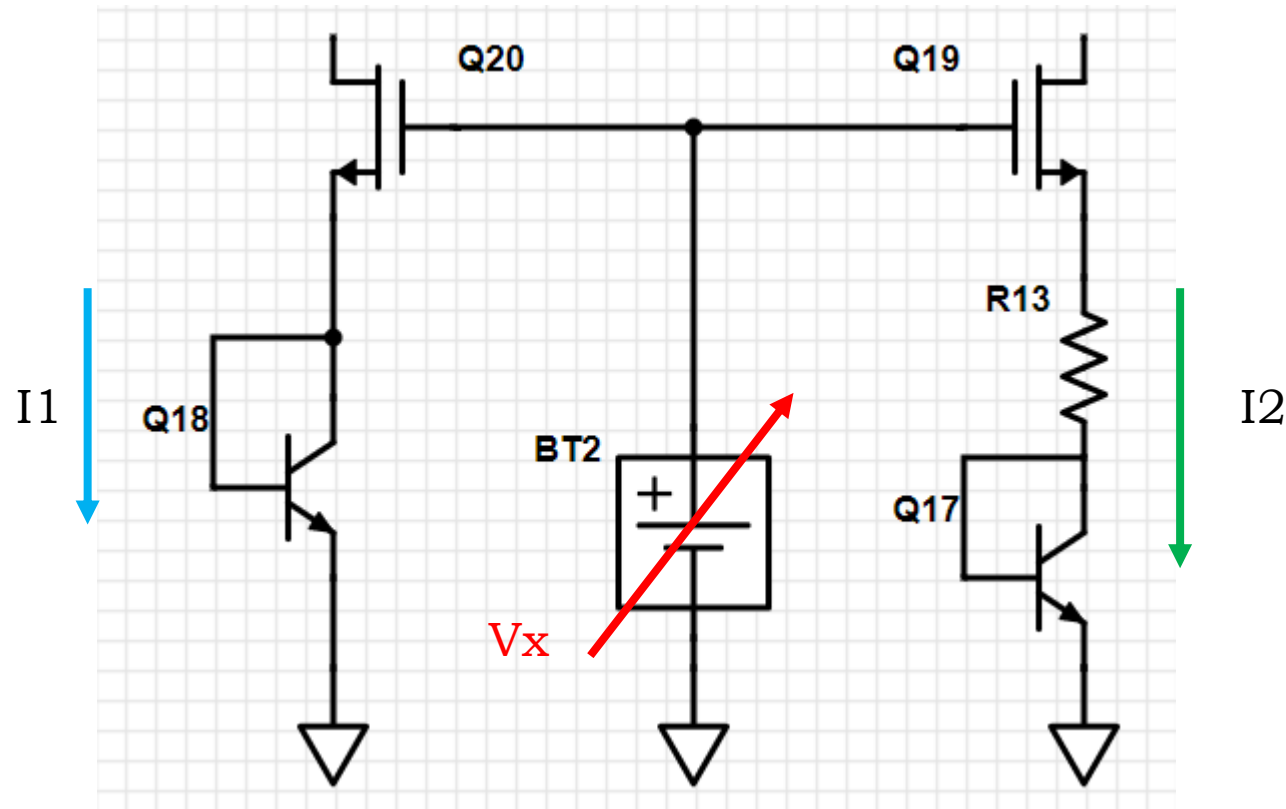
$I_1 < I_2$ ----- V_x is small ----- V_x needs to be increased !!!

$I_2 < I_1$ ----- V_x is large ----- V_x needs to be decreased !!!

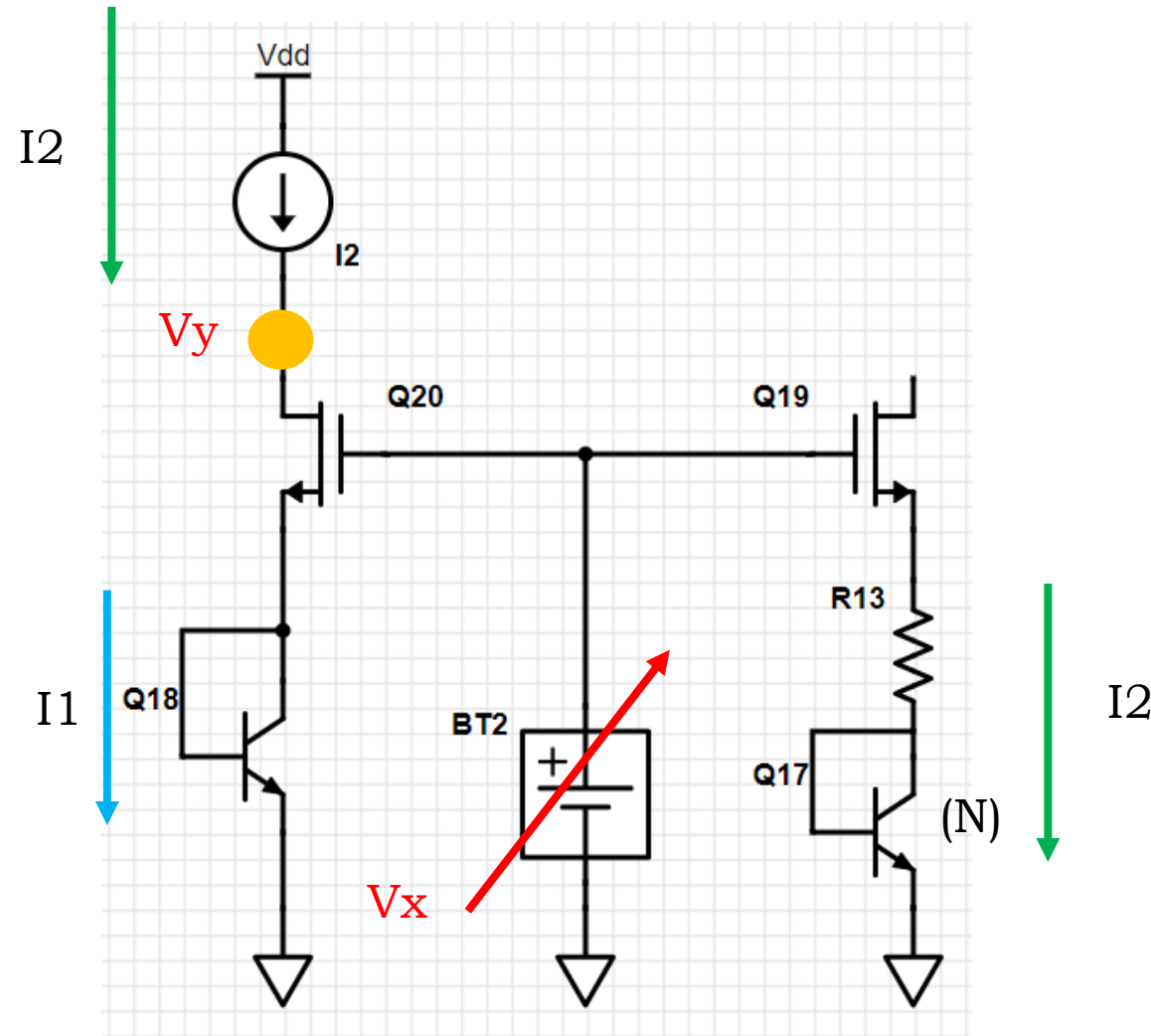
Basic design concept realization



Basic design concept realization

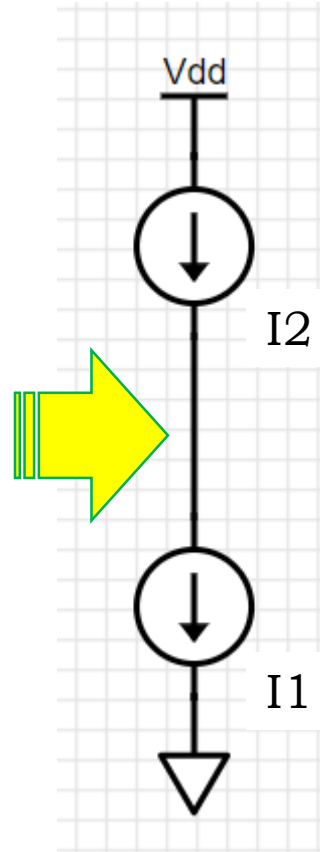


Basic design concept realization



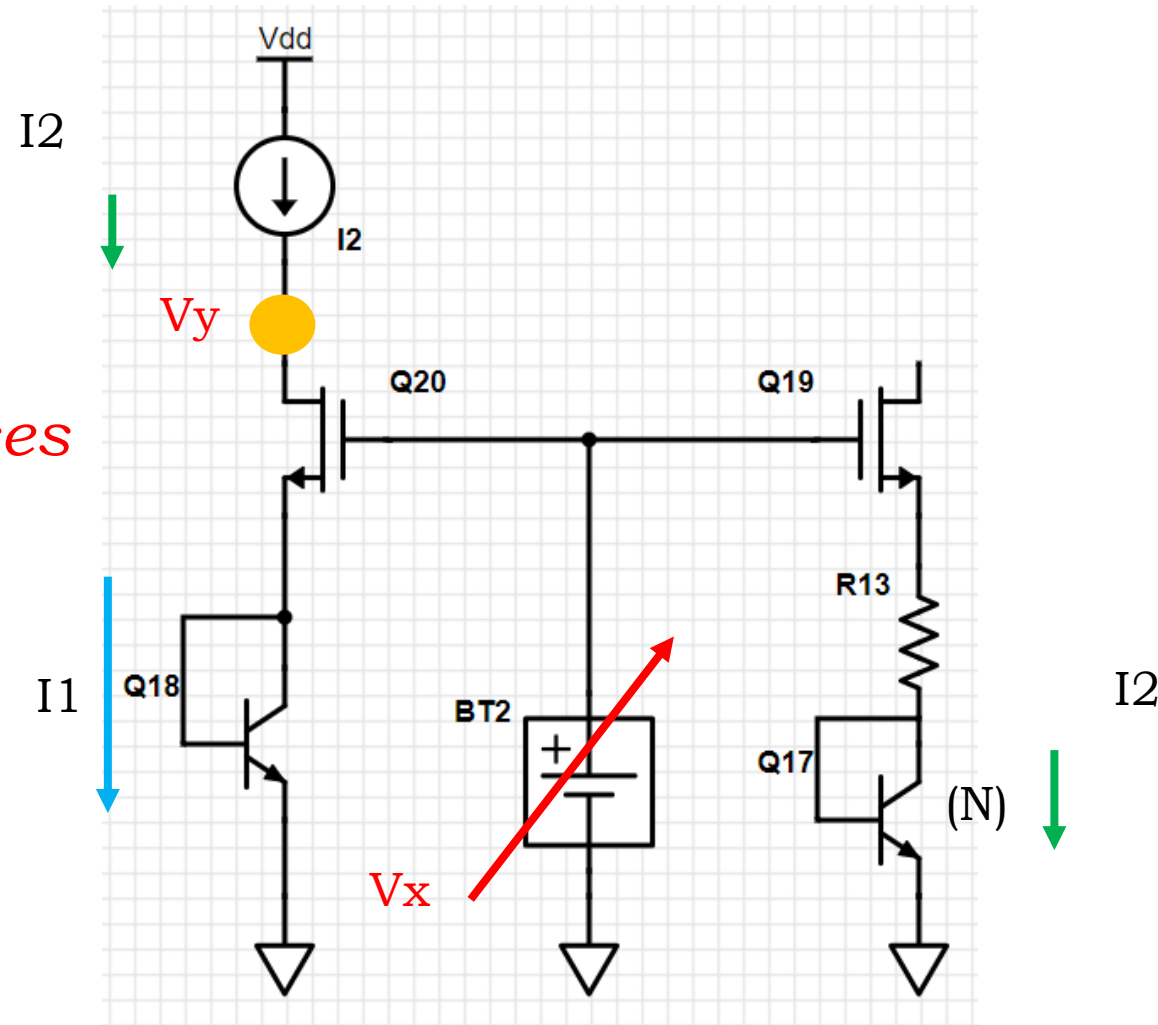
Basic design concept realization

Middle Potential ???



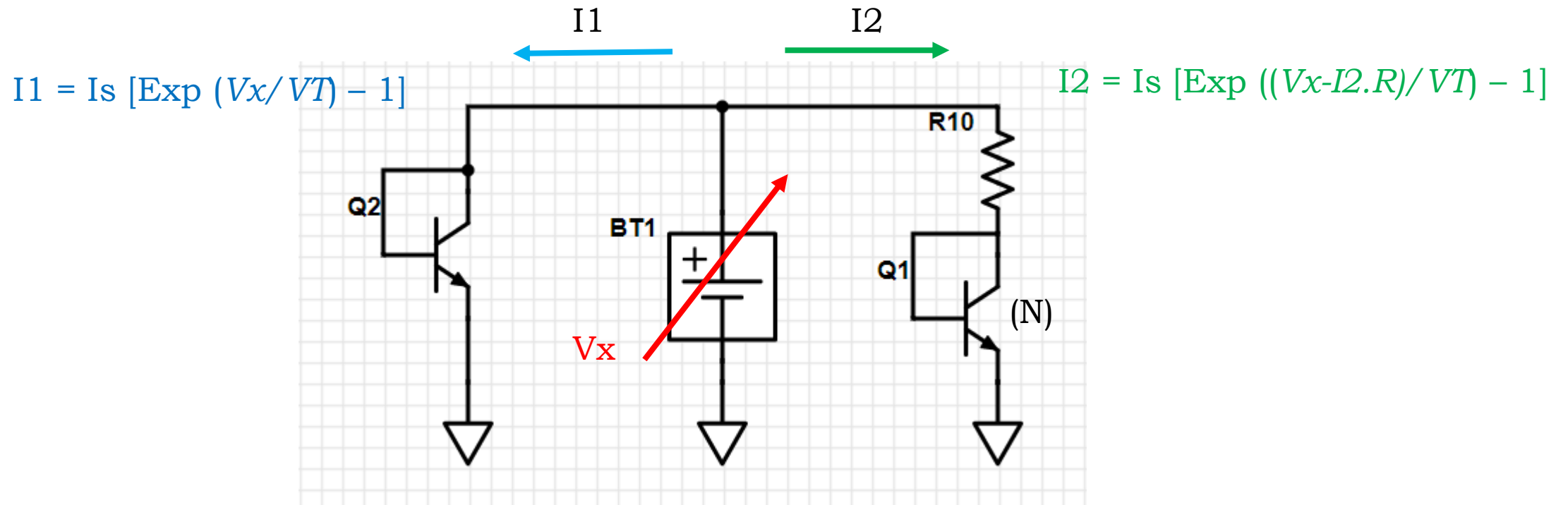
Basic design concept realization

$I_2 < I_1$ --- V_y decreases



Basic design concept realization

We have to make $I_1 = I_2 = I = \Delta V_{BE}/R$

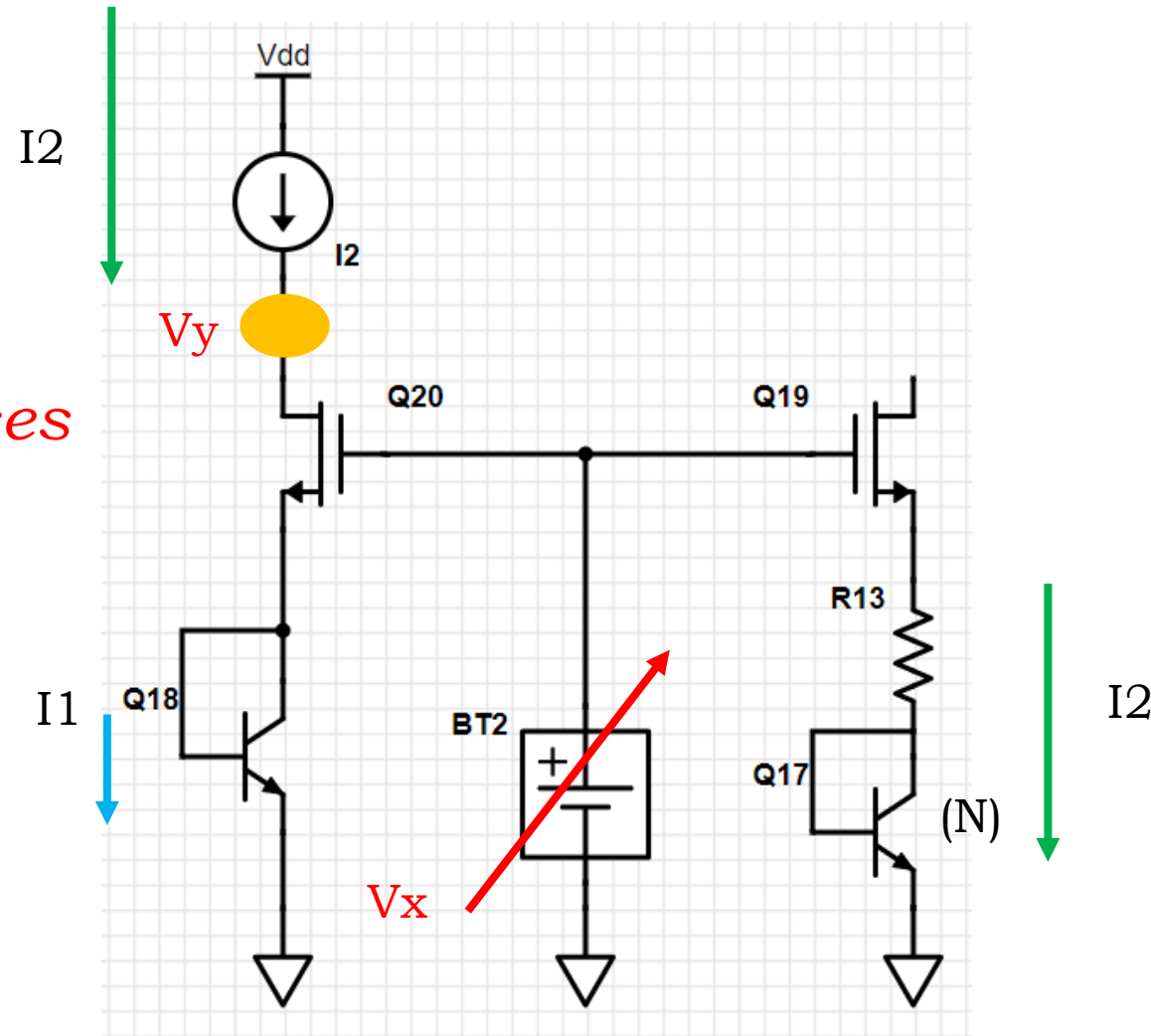


$I_1 < I_2$ ----- V_x is small ----- V_x needs to be increased !!!

$I_2 < I_1$ ----- V_x is large ----- **V_x needs to be decreased !!!**

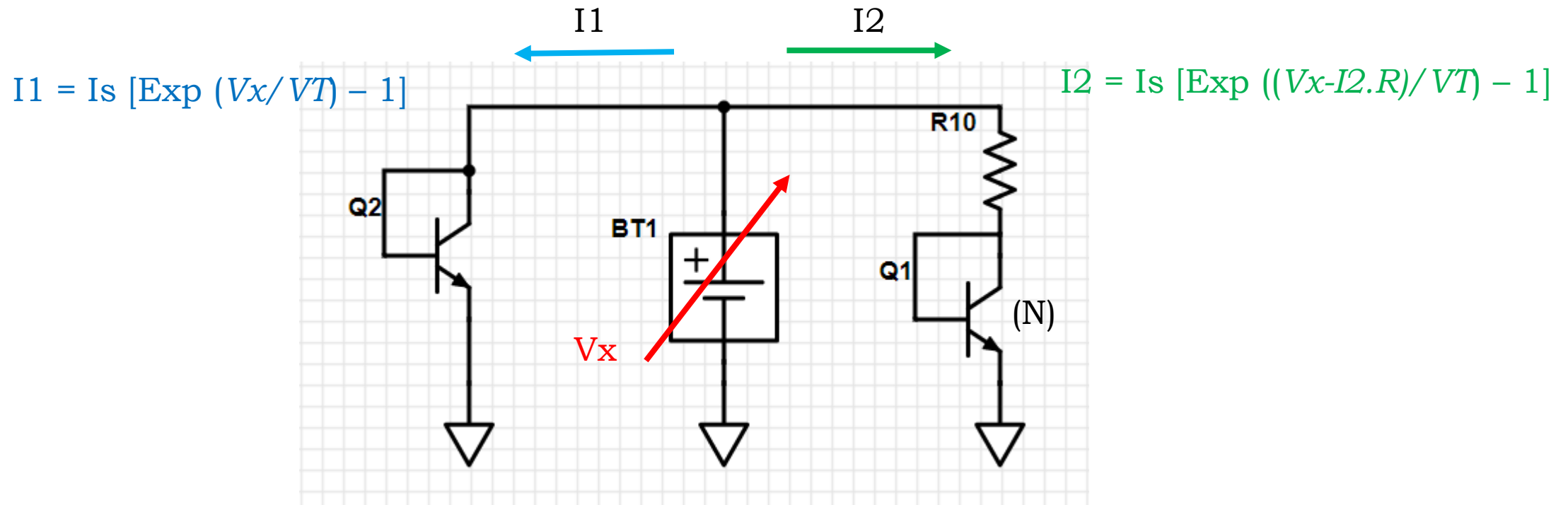
Basic design concept realization

$I_1 < I_2$ --- V_y increases



Basic design concept realization

We have to make $I_1 = I_2 = I = \Delta V_{BE}/R$



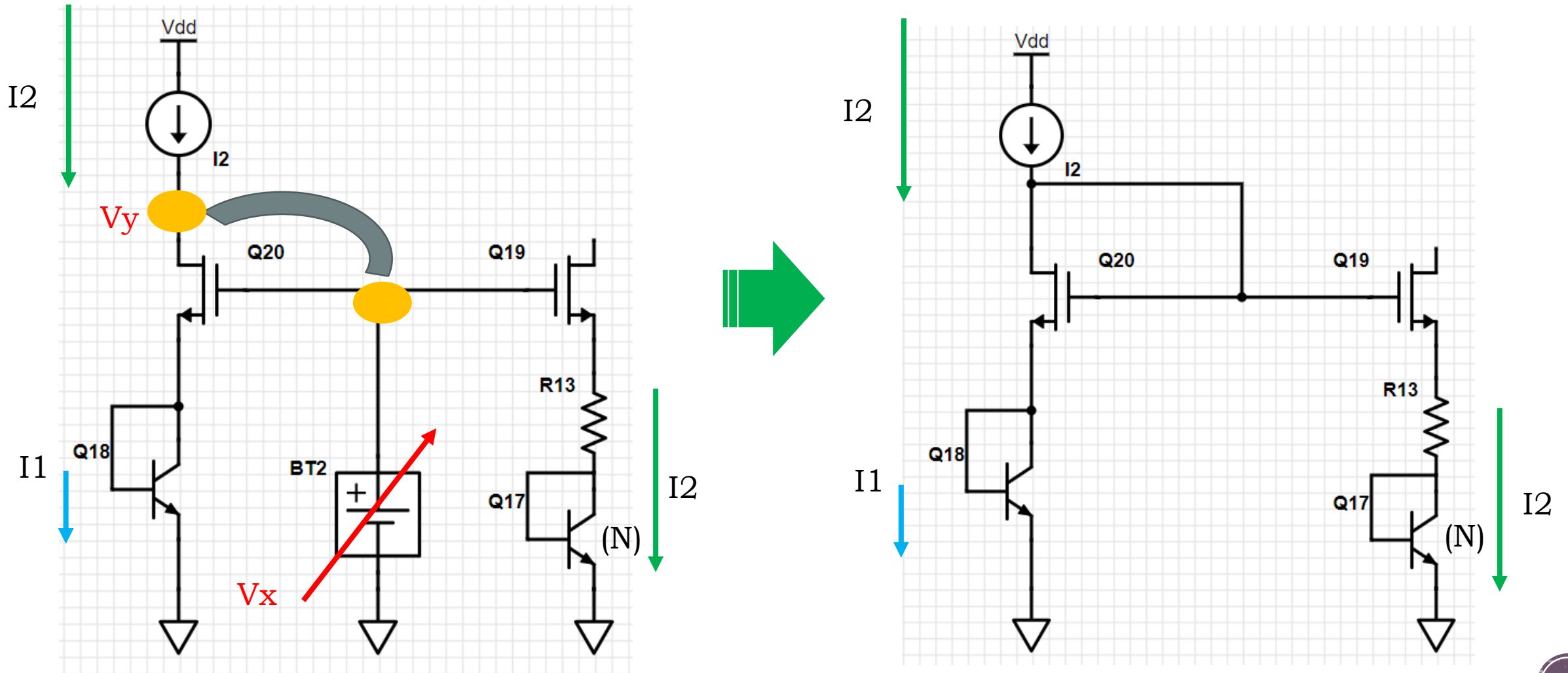
$I_1 < I_2$ ----- V_x is small ----- **V_x needs to be increased !!!**

$I_2 < I_1$ ----- V_x is large ----- V_x needs to be decreased !!!

Basic design concept realization

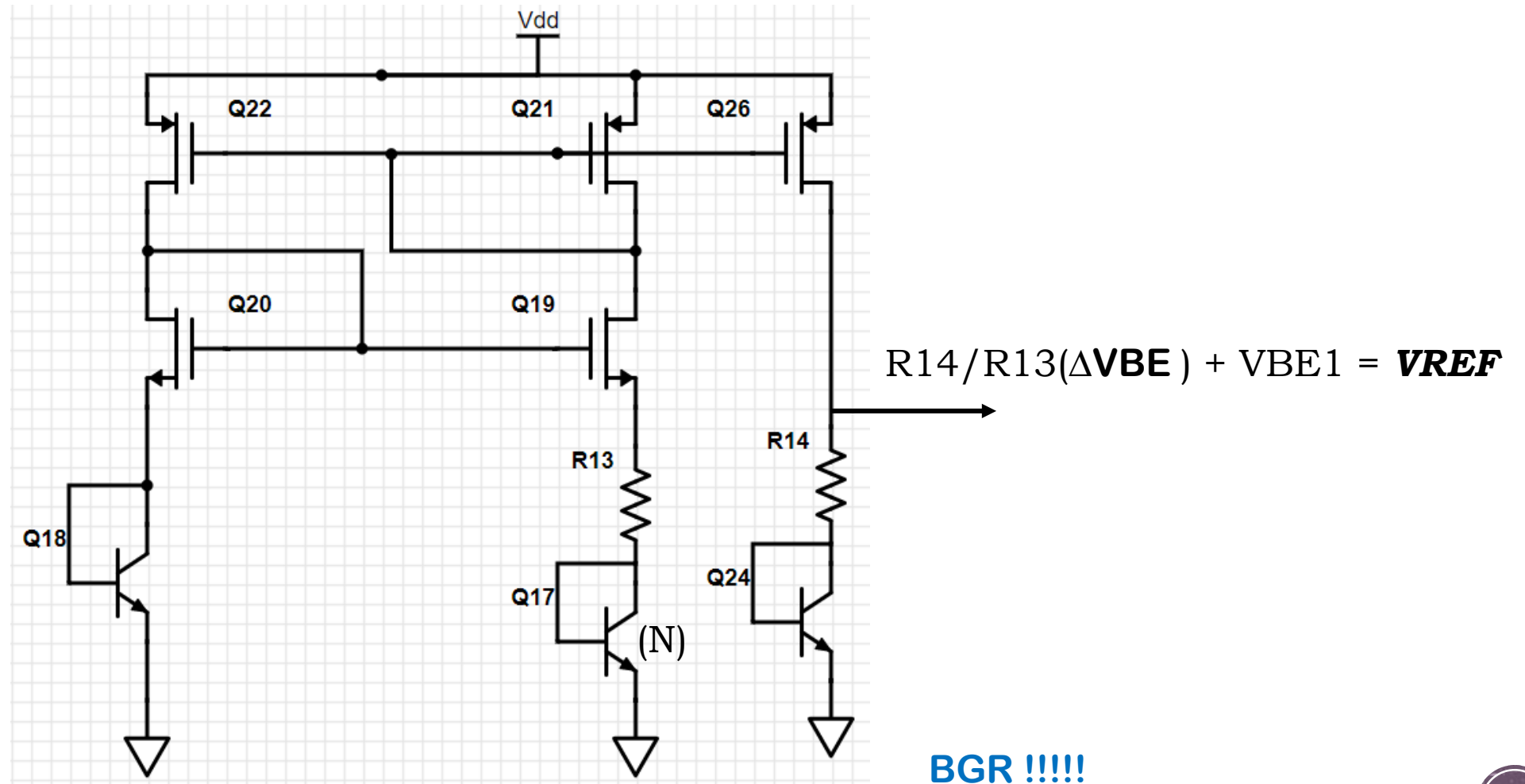
V_x and V_y are moving in same direction

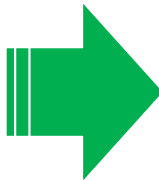
Basic design concept realization





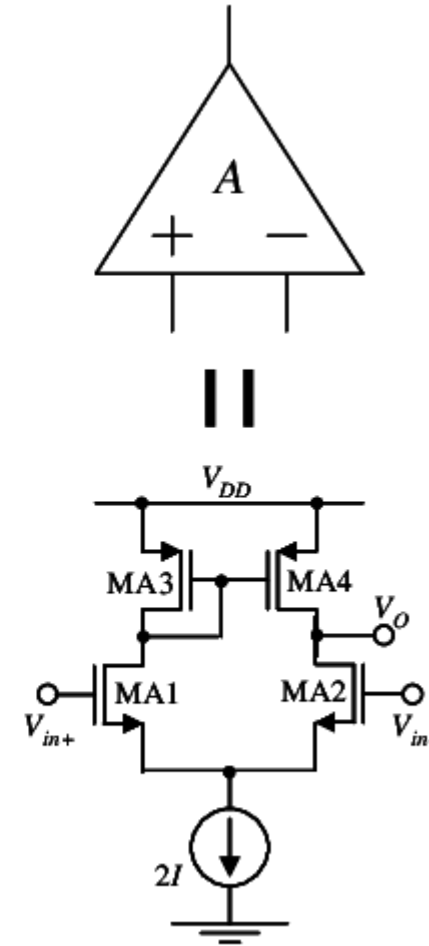
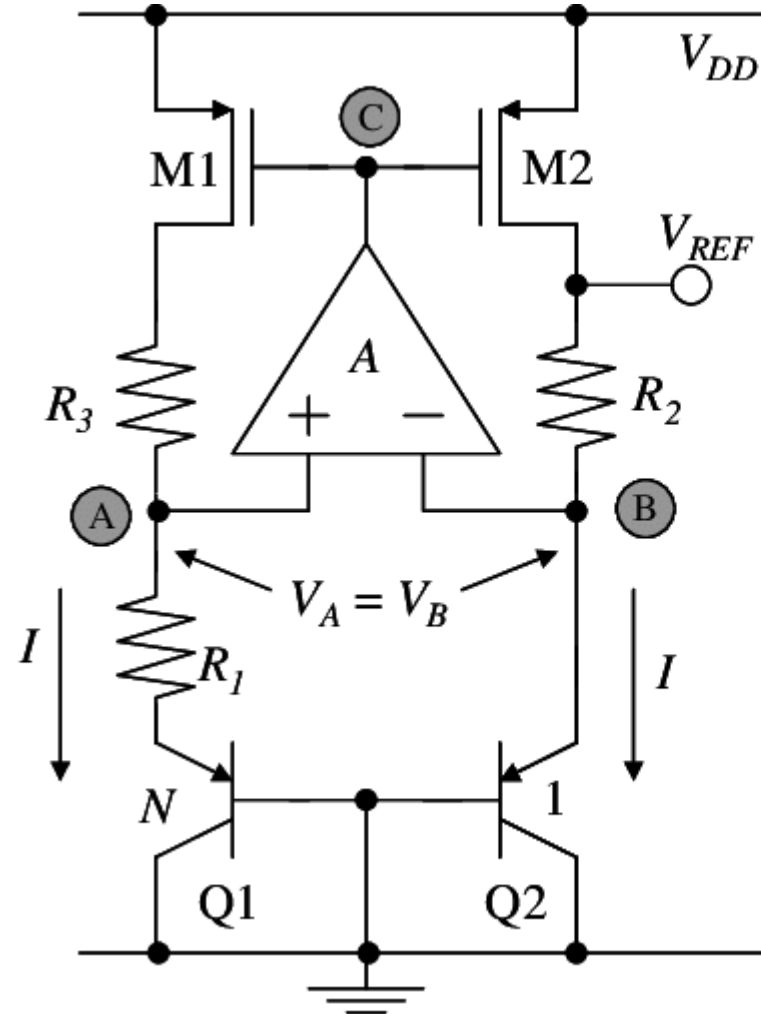
Basic design concept realization

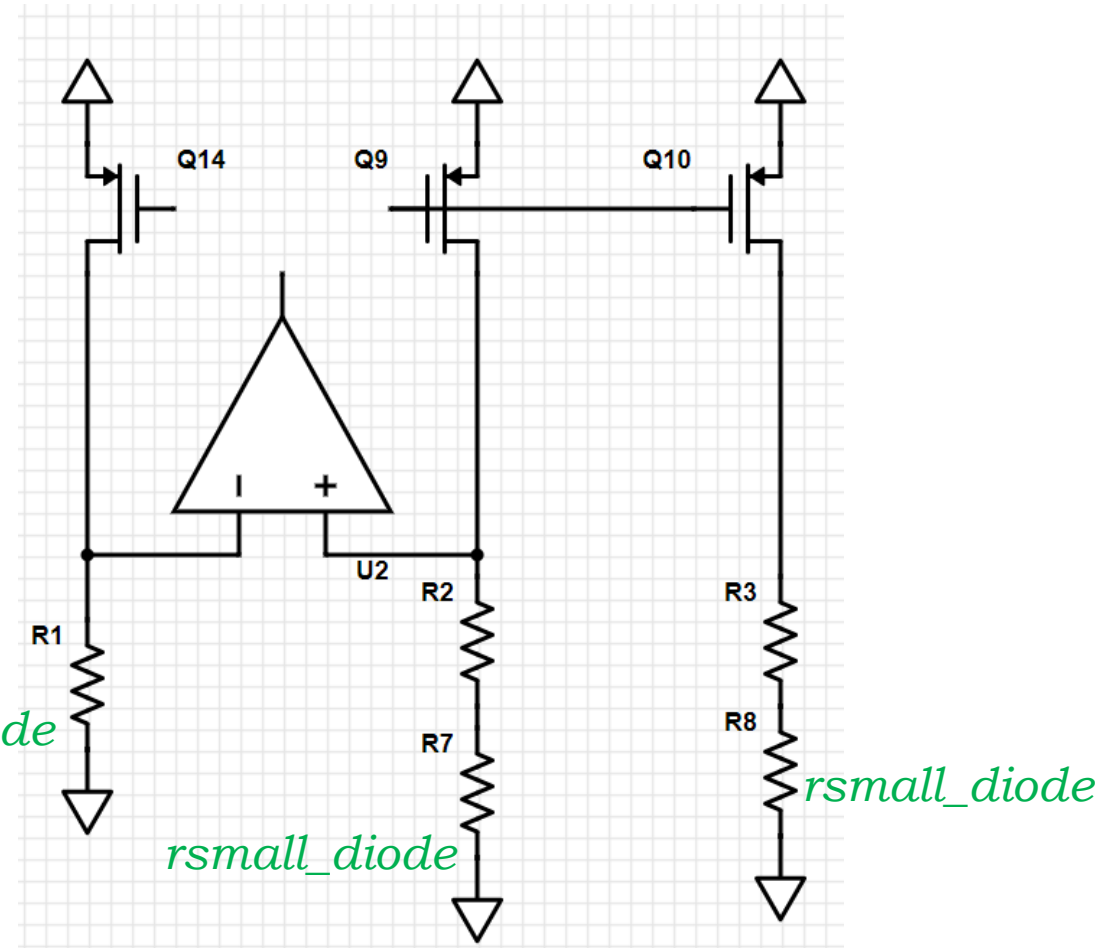
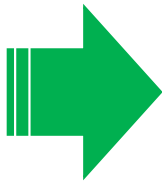




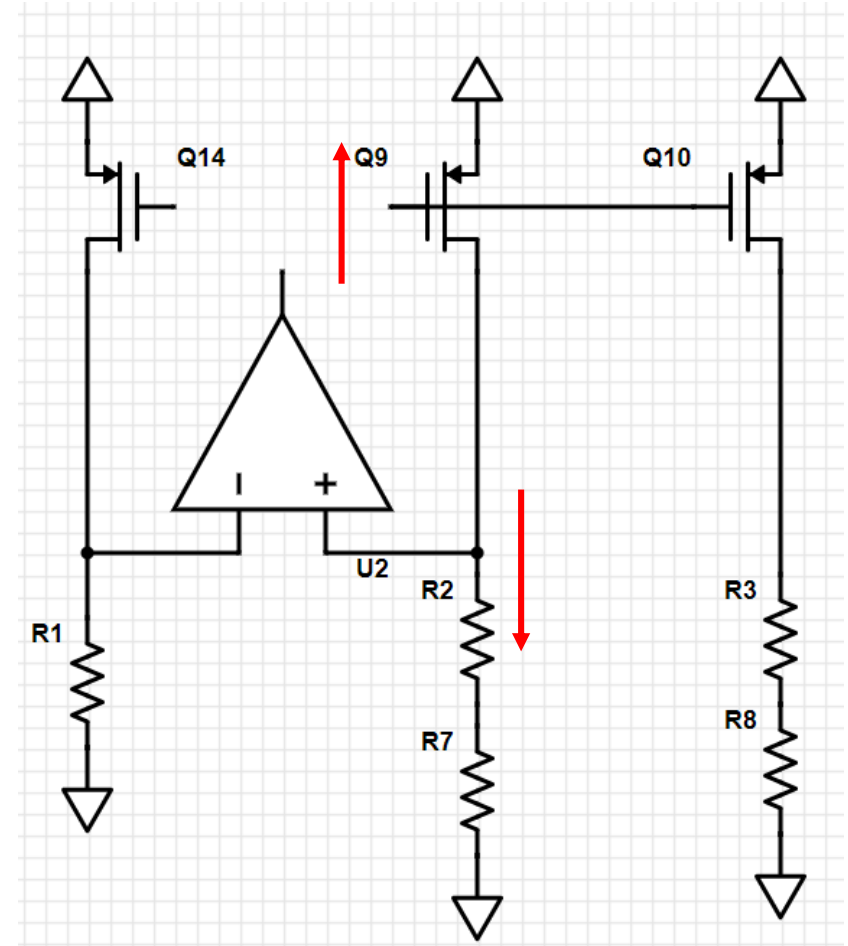
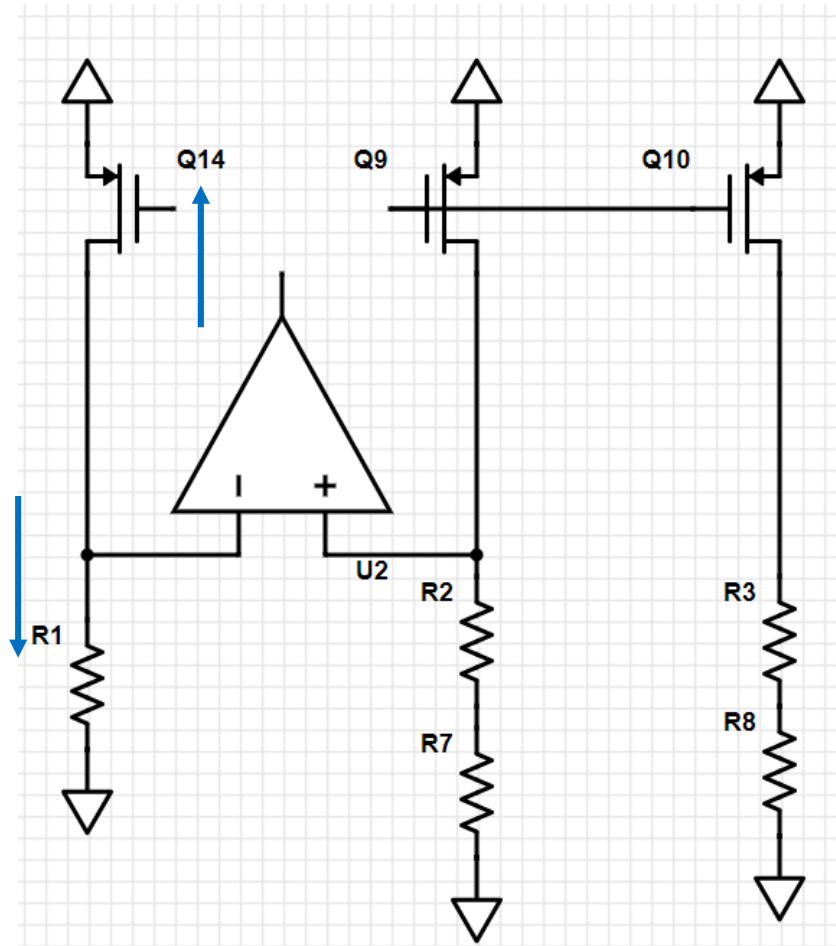
Things to remember!!

Simple OPAMP

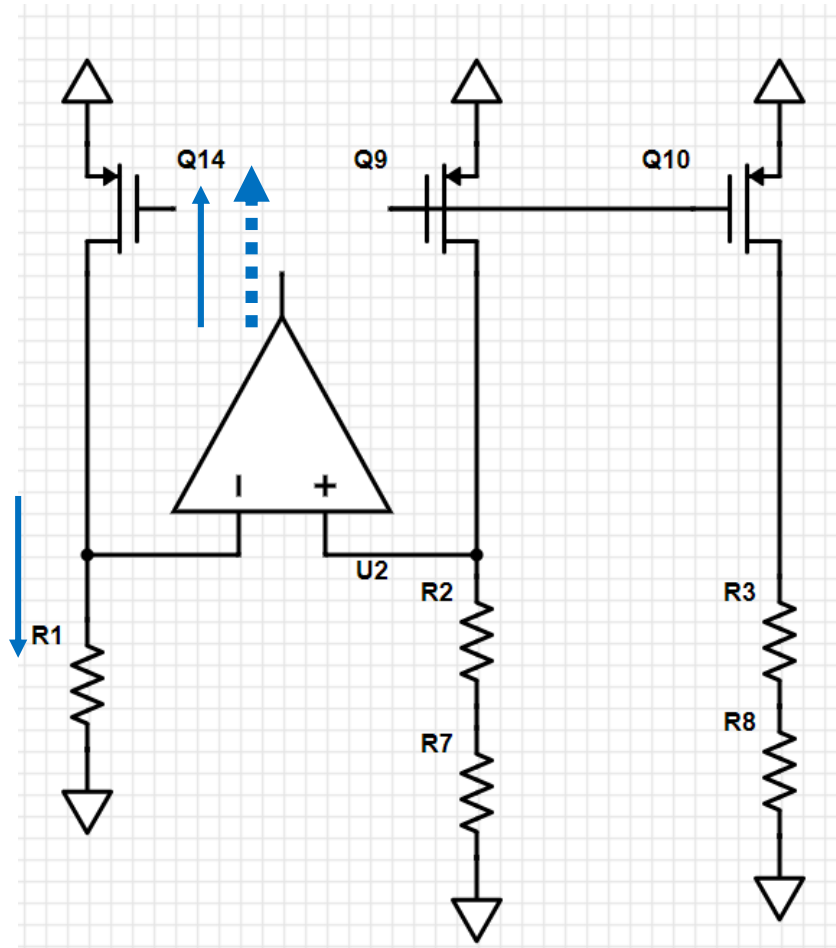




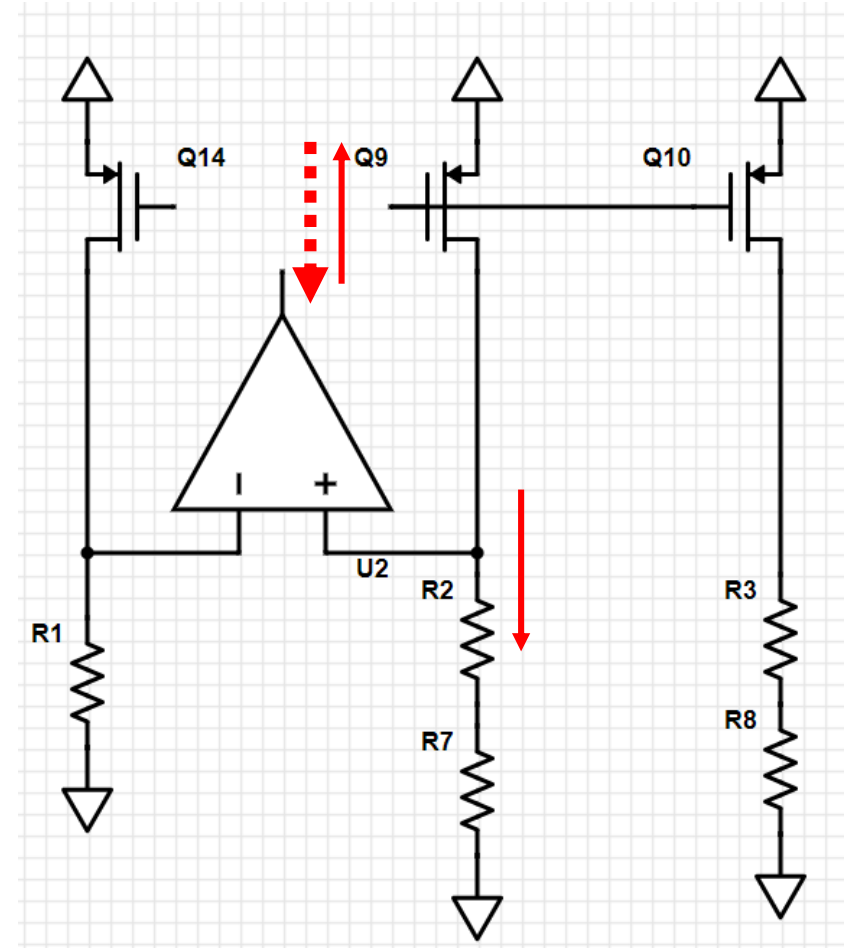
Small Signal Analysis



Small Signal Analysis

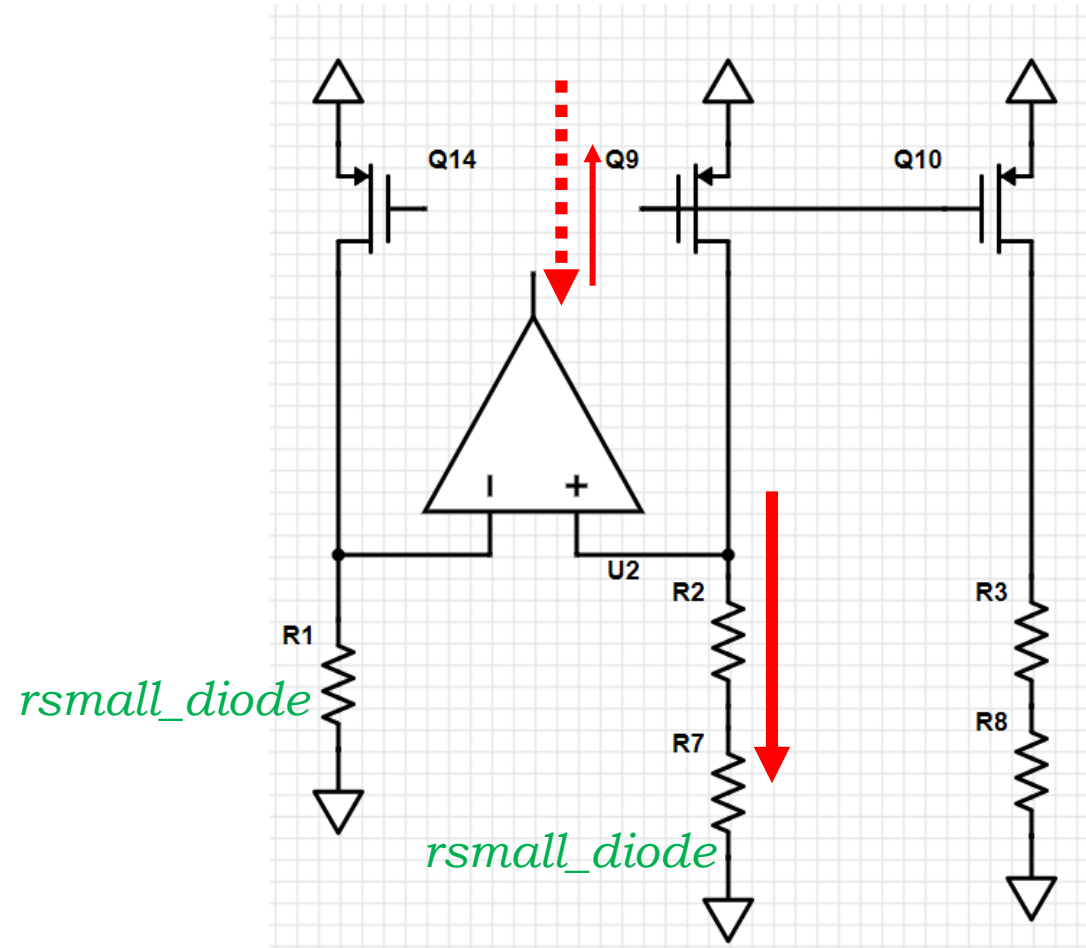
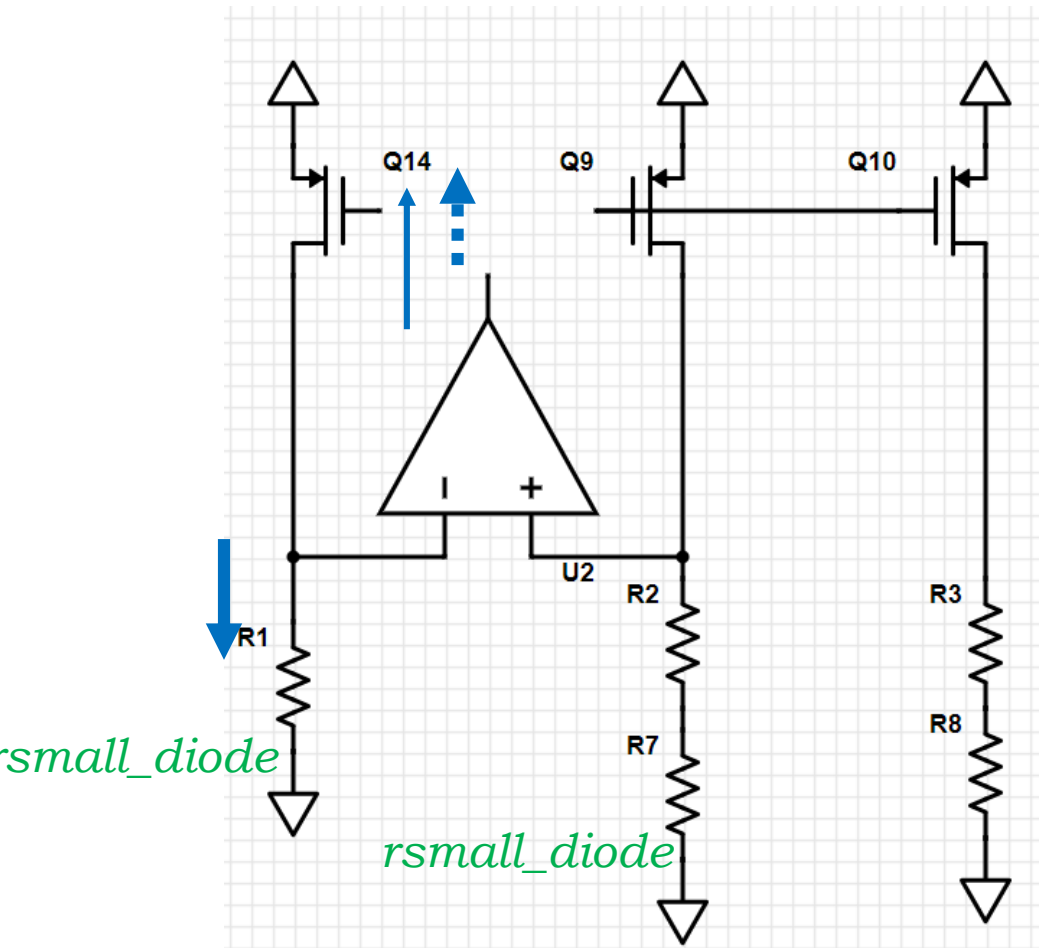


Positive Feedback



Negative Feedback

Small Signal Analysis

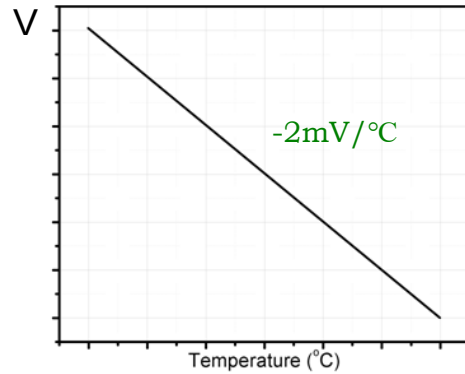


Positive Feedback **WEAKER than** Negative Feedback

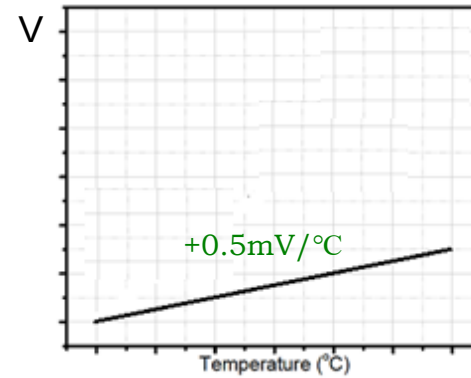
Case Study

Basics of voltage reference

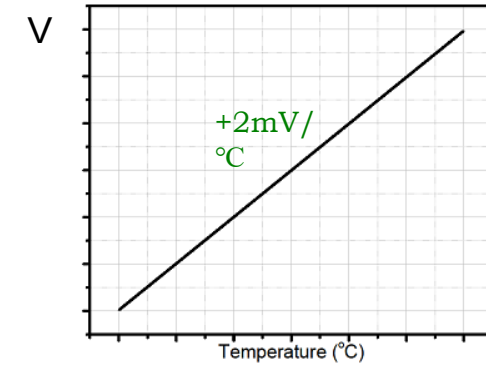
1. CTAT Voltage



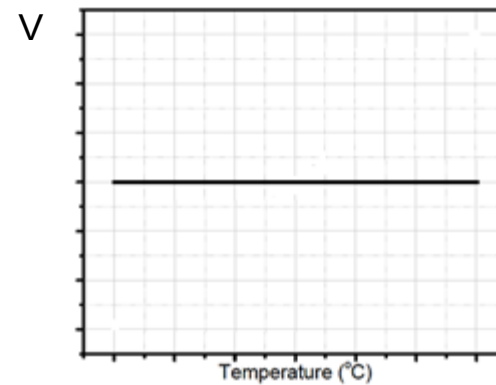
2. PTAT Voltage



3. Scaled PTAT

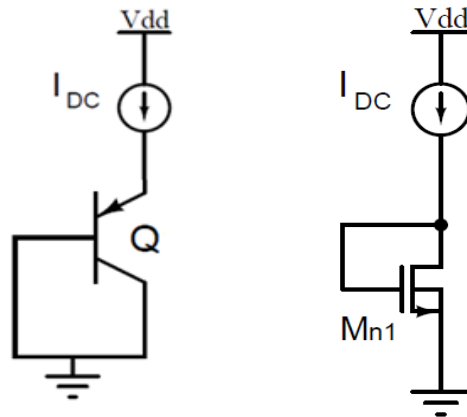
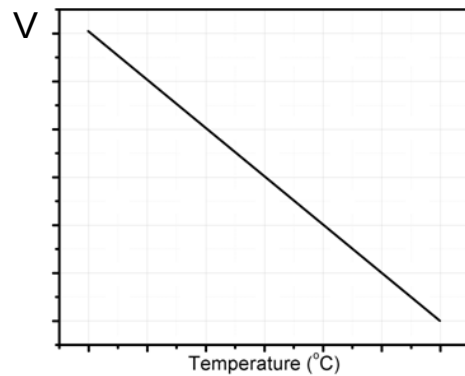


4. Adding (1 and 3)



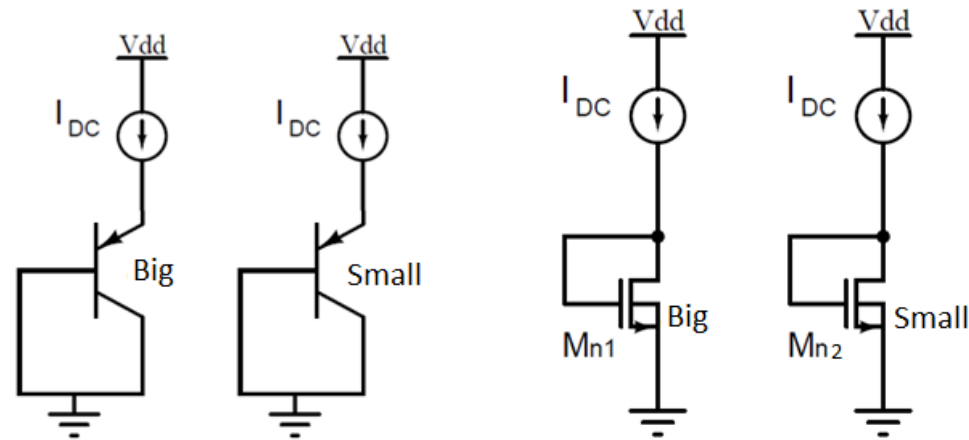
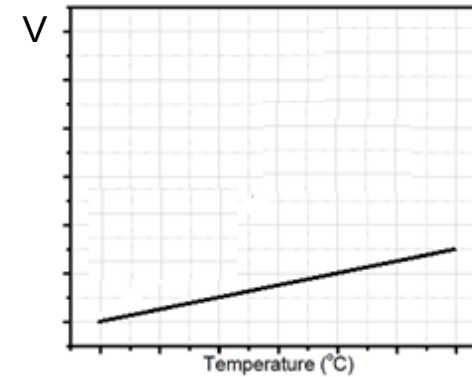
Basics of voltage reference

1. CTAT Voltage



V_{EB} , V_{GS} is a CTAT voltage

2. PTAT Voltage



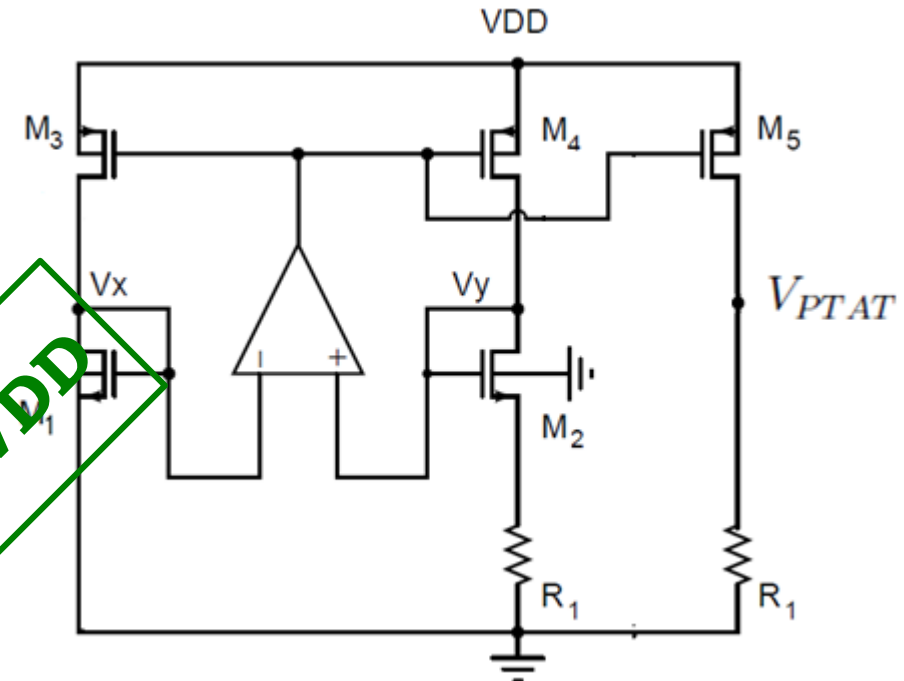
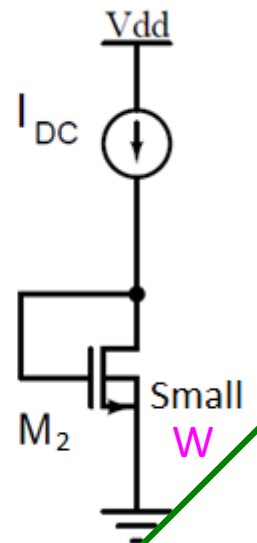
ΔV_{EB} , ΔV_{GS} is a PTAT voltage

PTAT voltage generator



Small W

Process & VDD

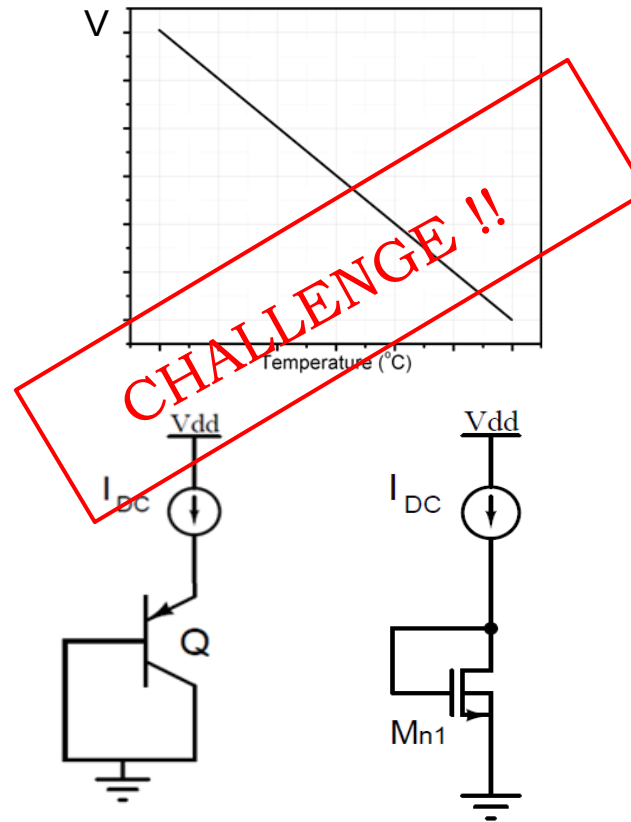
$$V_{PTAT} = V_T \ln \left(\frac{I_1}{I_2} \right)$$


$$V_{PTAT} = V_T \ln (M)$$

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Basics of voltage reference

CTAT Voltage

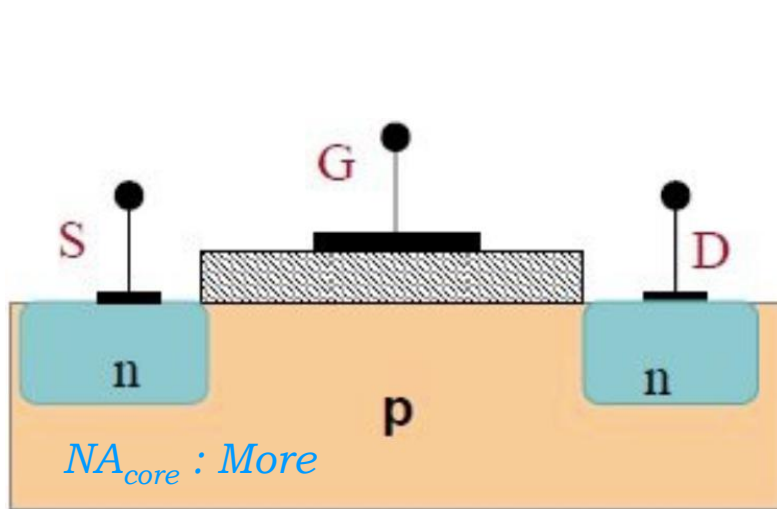


V_{EB} , V_{GS} is a CTAT voltage

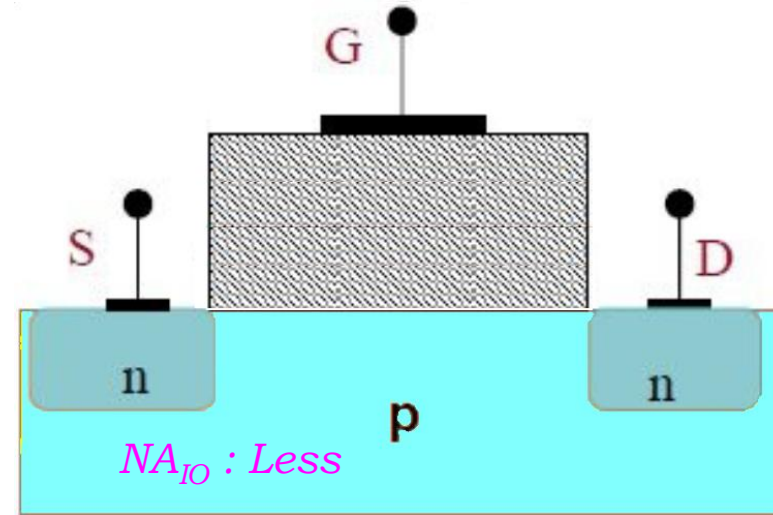
V_{GS} is **Heavily** Process Variant

1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.1 V_{REF} using ΔV_{TH} of core-IO MOS pair



$$\begin{aligned}\Delta V_{GS} &= V_{GS_{IO}} - V_{GS_{core}} \\ &= \Delta V_{TH}\end{aligned}$$



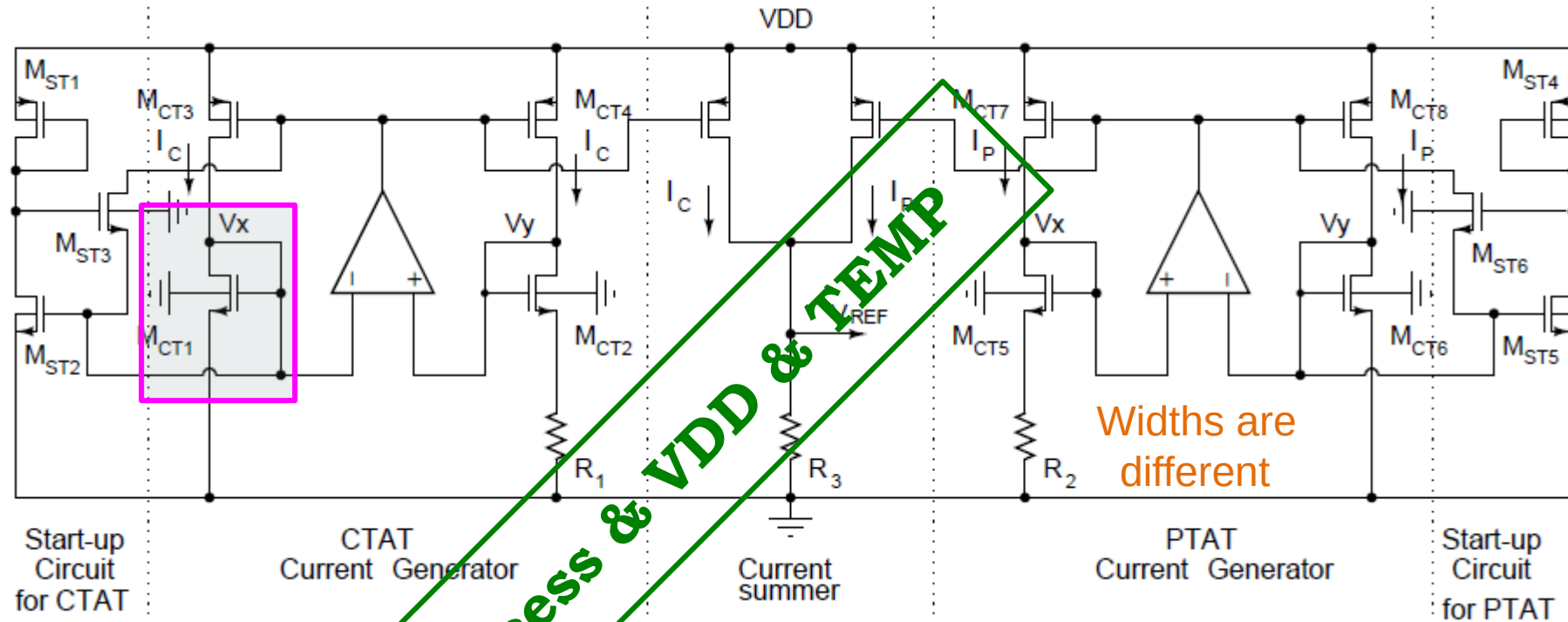
$$\frac{d\Delta V_{GS}}{dT} = \frac{2k}{q} \left[\ln \frac{N_{A_{IO}}}{N_{A_{core}}} \right]$$

Difference **POSITIVE** and Slope **NEGATIVE** !!!

Higher substrate doping concentration : Results in **increasing V_{TH}**

1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.1 V_{REF} using ΔV_{TH} of core-IO MOS pair



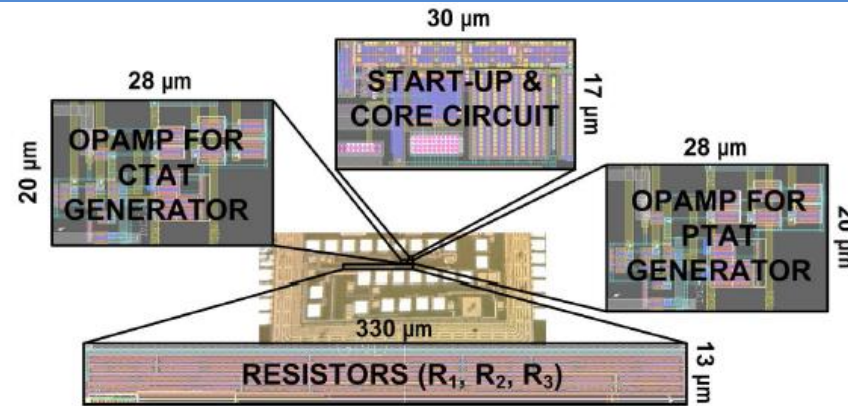
$$I_{CTAT} = \frac{\Delta V_{TH}}{R_1}$$

$$I_{PTAT} = \frac{V_T}{R_2} \ln(M)$$

$$V_{REF} = \Delta V_{TH} + \frac{R_1}{R_2} V_T \ln(M)$$

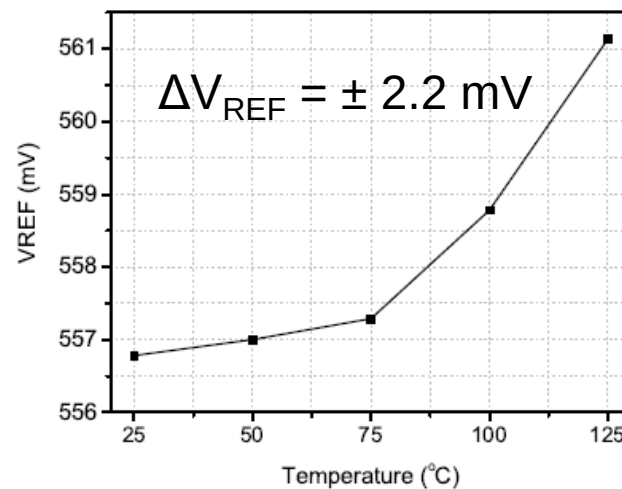
1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.1 V_{REF} using ΔV_{TH} of core-IO MOS pair

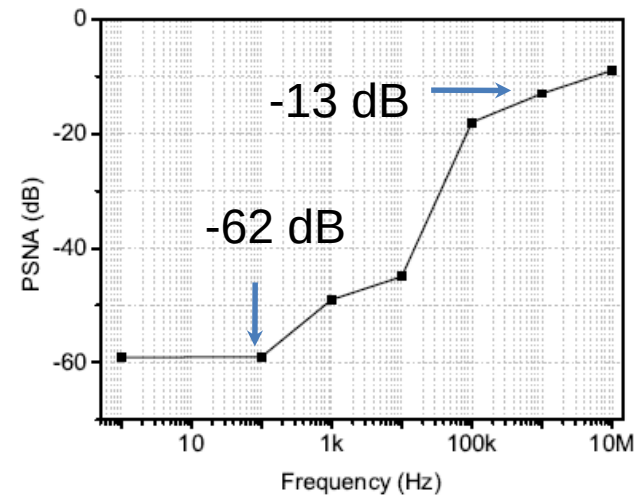


Area = 0.028 mm²

Chip micrograph



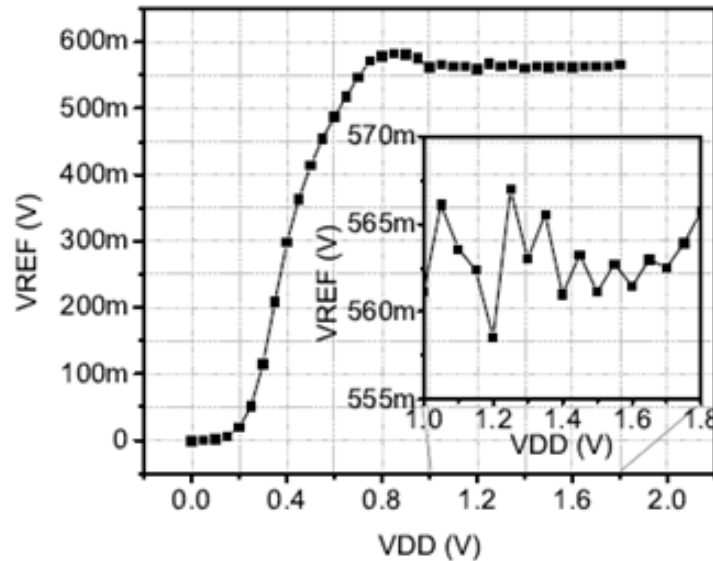
Reference voltage with temperature



PSNA response

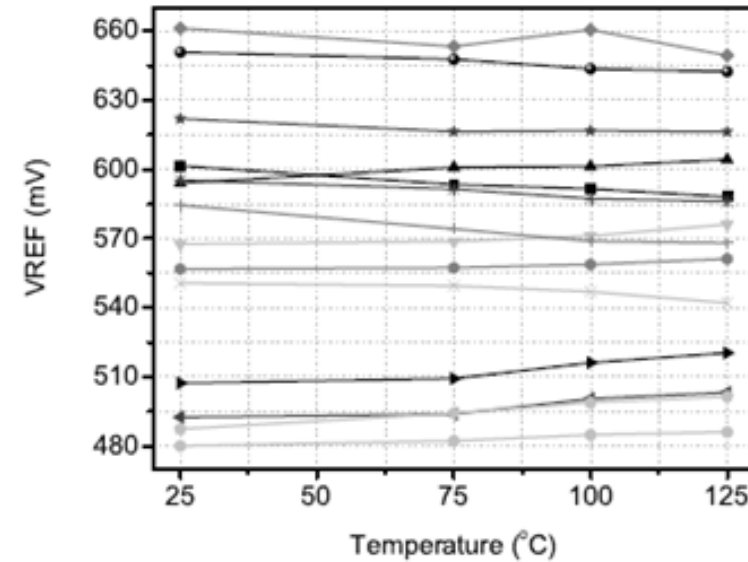
1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.1 V_{REF} using ΔV_{TH} of core-IO MOS pair



Line sensitivity

Line sensitivity = ± 5 mV/V
[$V_{DD} = 1\text{ V} : 1.8\text{ V}$]



Reference voltage with temperature
for 14 samples

V_{REF} spread = ± 90 mV (14 samples)

1. V_{REF} by linear combination of ΔV_{TH} and V_T

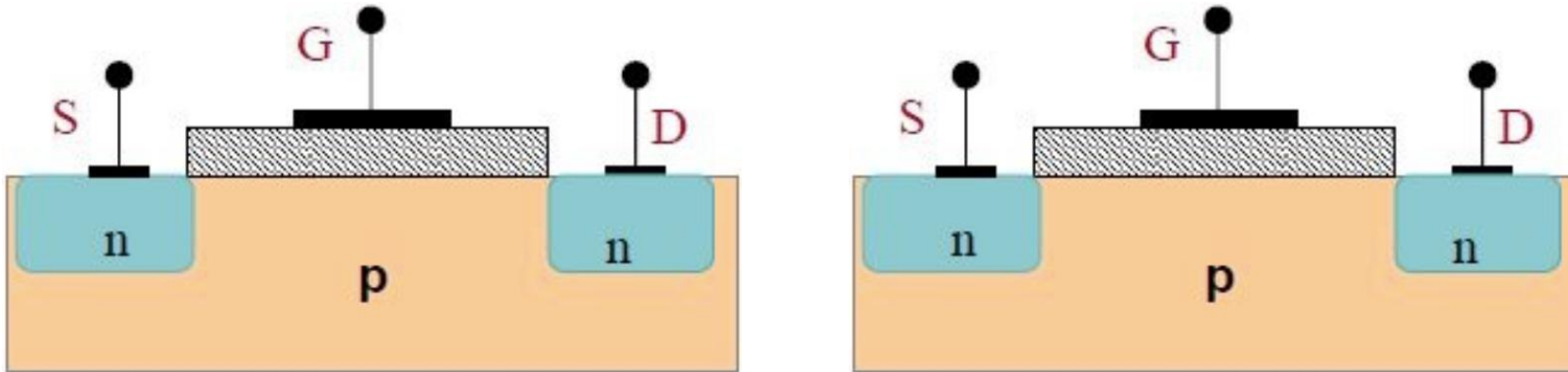
1.1 V_{REF} using ΔV_{TH} of core-IO MOS pair

Performance of proposed voltage reference circuit

Architecture	$\Delta V_{TH} - V_T$ Core-IO (Measured)
Technology	180 nm
V_{DD} (V)	1
I_{DD} (A)	15 μ
Power Cons. (W)	15 μ
ΔV_{REF} (V)	± 2.2 m
MAPV (%)	0.79
Temp. Range ($^{\circ}$ C)	25 : 125
Area (mm ²)	0.028
Voltage Spread (V)	± 90 m

1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.2 V_{REF} using ΔV_{TH} of core-core MOS pair

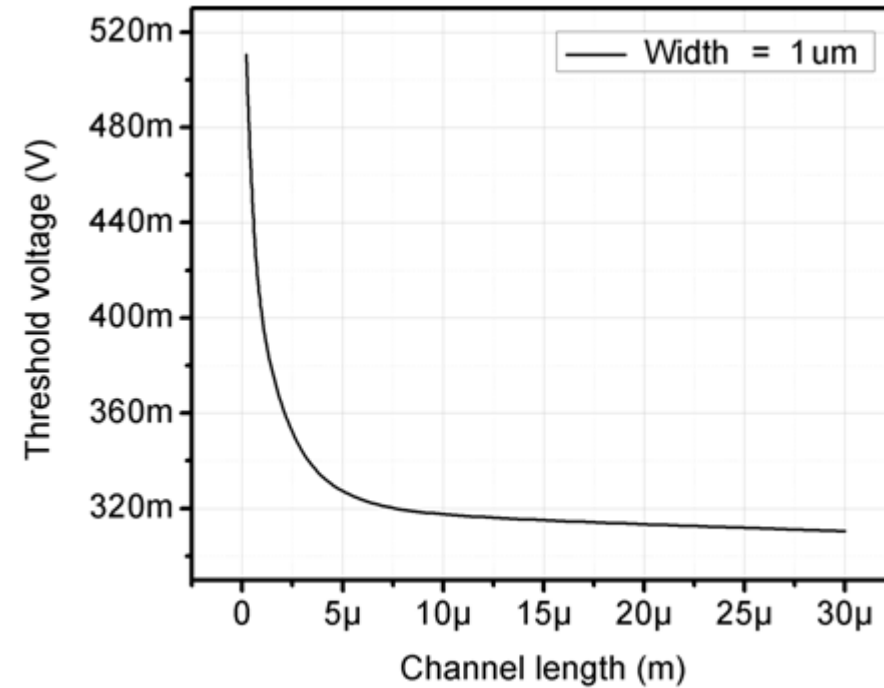
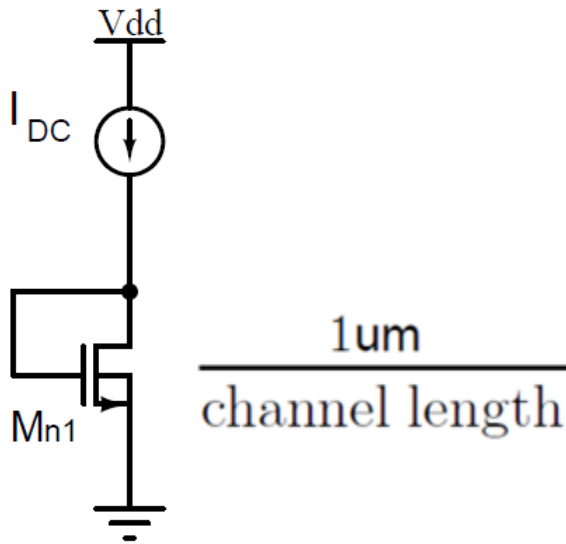


We need to use only 'CORE' MOSFETs to avoid process variations.

1. V_{REF} by linear combination of ΔV_{TH} and V_T

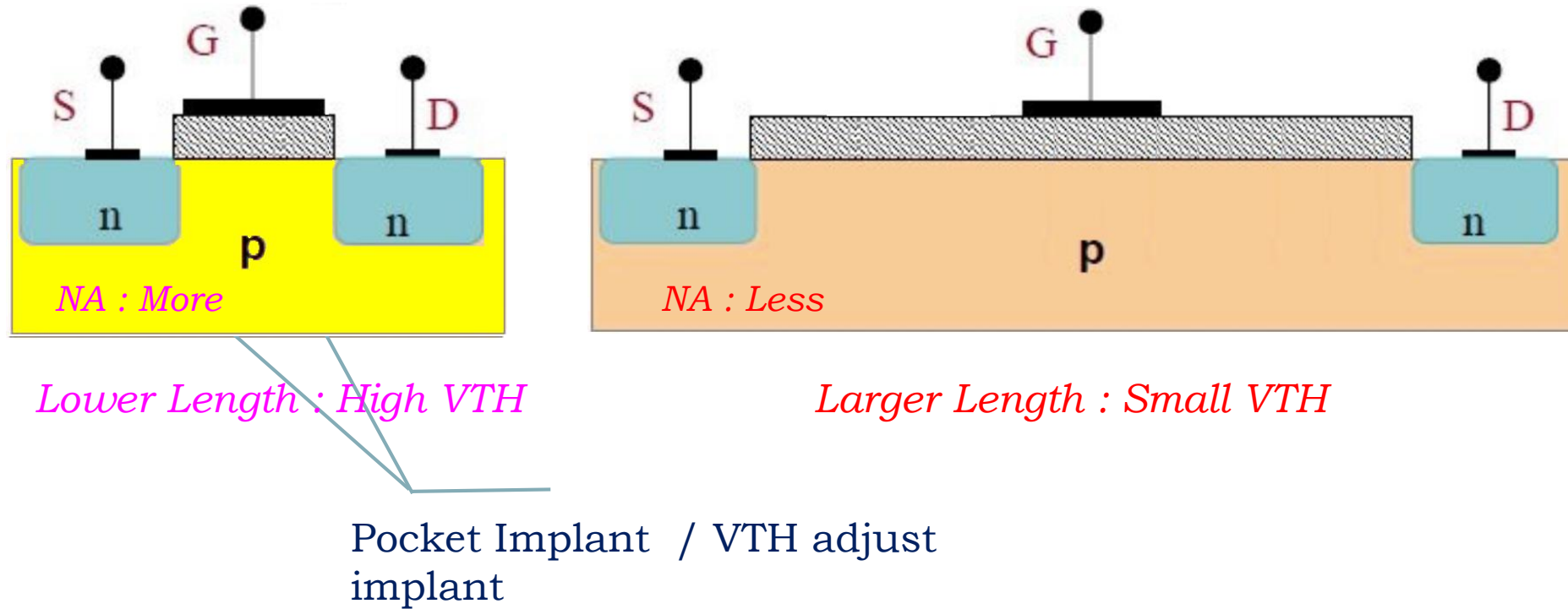
1.2 V_{REF} using ΔV_{TH} of core-core MOS pair

Reverse short channel effect :



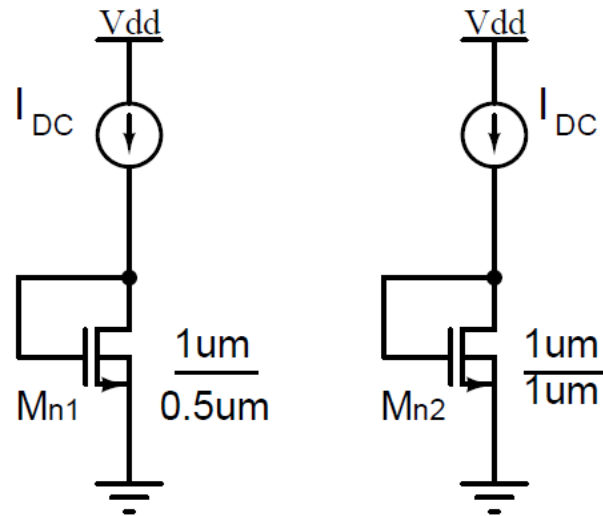
1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.2 V_{REF} using ΔV_{TH} of core-core MOS pair



1. V_{REF} by linear combination of ΔV_{TH} and V_T

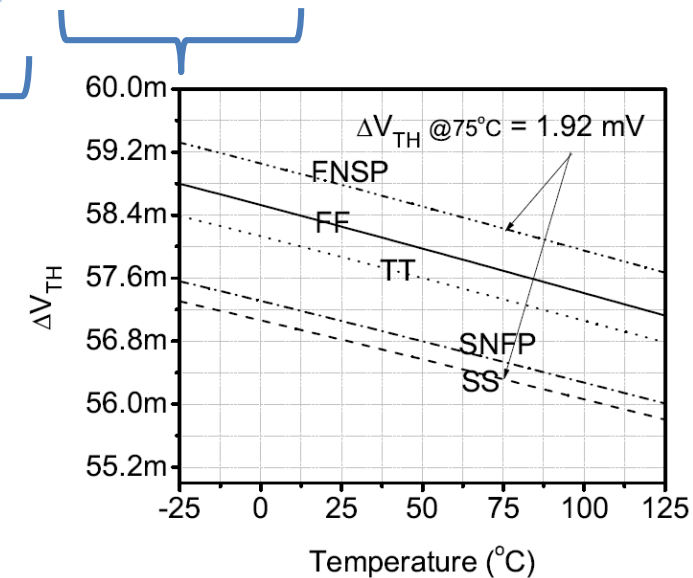
1.2 V_{REF} using ΔV_{TH} of core-core MOS pair



$N = \text{Ratio of channel lengths}$

$$\Delta V_{GS} = V_T \ln \left[\frac{1}{N} \right] + \Delta V_{th}$$

$N = 2$

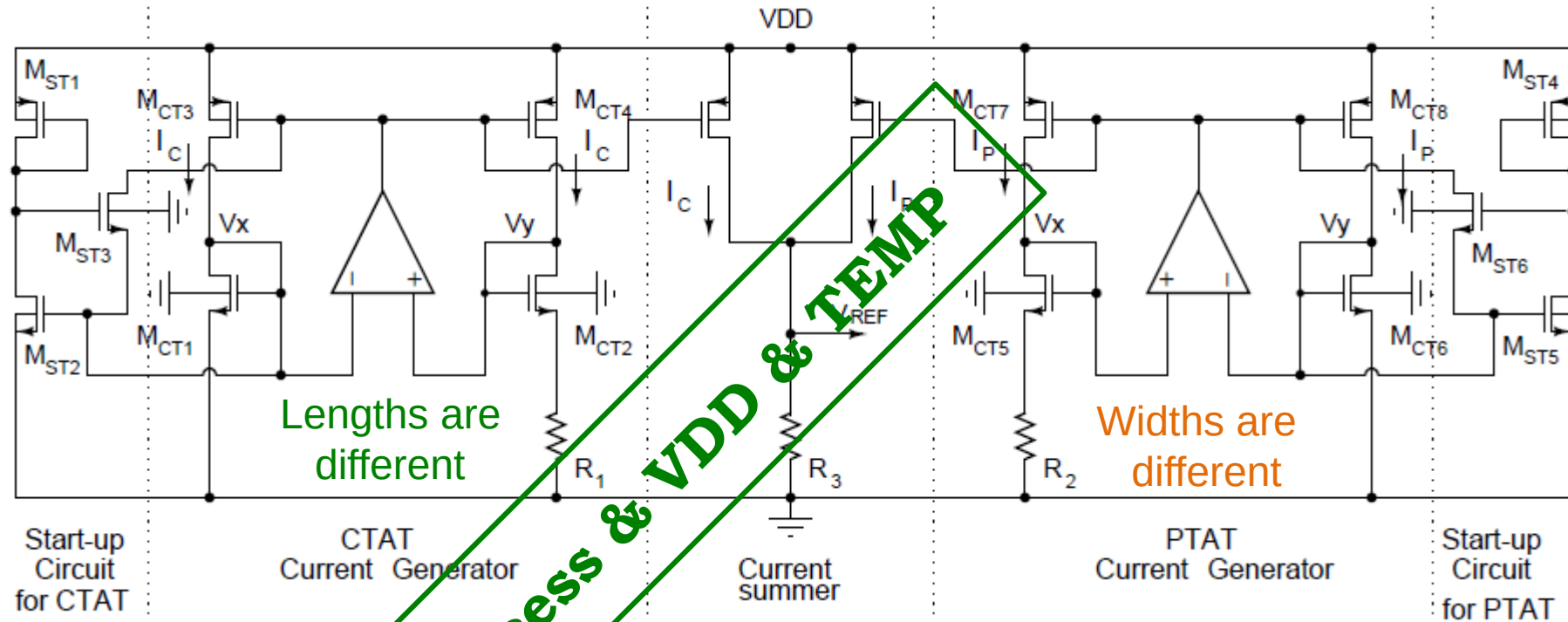


ΔV_{GS} shows a CTAT behavior because

1. Magnitude is positive as value of $N > 1$ (upper limit is decided by ΔV_{TH})
2. Slope of ΔV_{GS} is negative (both the terms has negative slope)

1. V_{REF} by linear combination of ΔV_{TH} and V_T

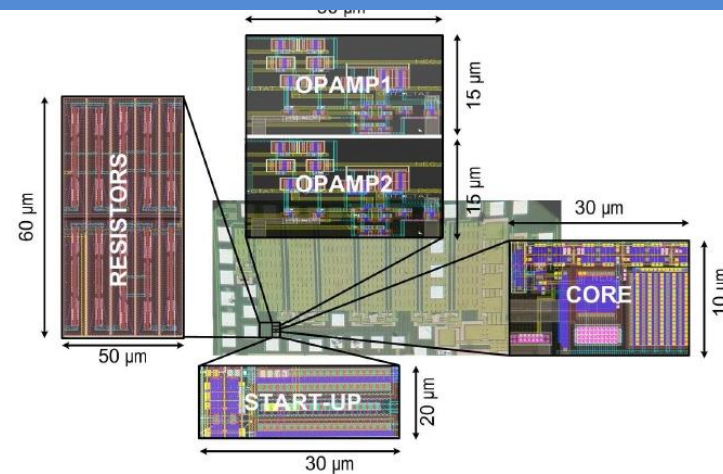
1.2 V_{REF} using ΔV_{TH} of core-core MOS pair



$$V_{REF} = V_T \ln\left(\frac{1}{N}\right) + \Delta V_{TH} + \frac{R_1}{R_2} V_T \ln(M)$$

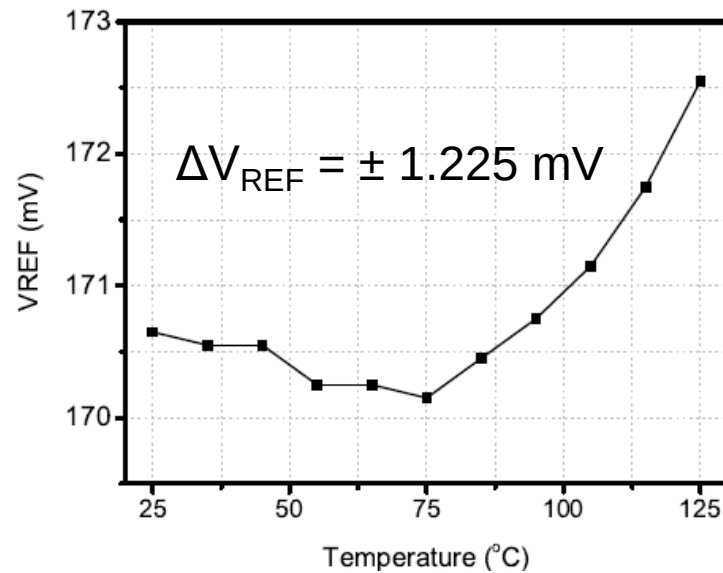
1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.2 V_{REF} using ΔV_{TH} of core-core MOS pair

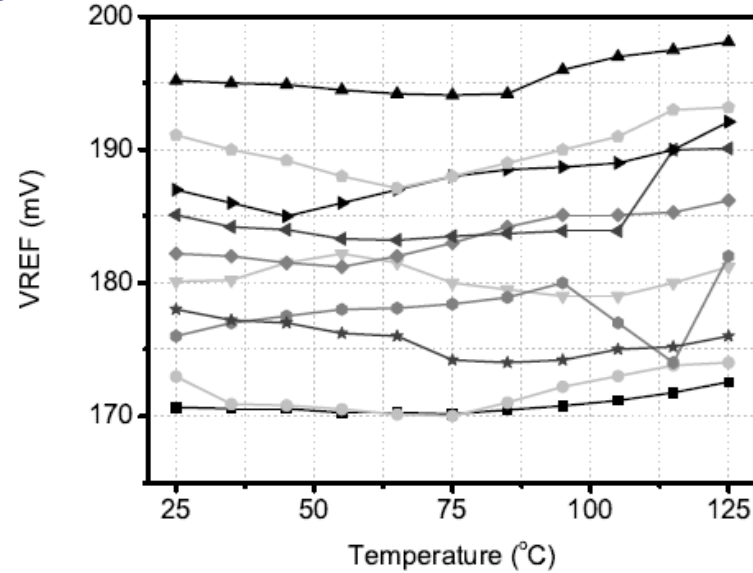


Area = 0.0048 mm²

Chip micrograph



Reference voltage with temperature



V_{REF} spread = ± 14 mV (10 samples)

1. V_{REF} by linear combination of ΔV_{TH} and V_T

1.2 V_{REF} using ΔV_{TH} of core-core MOS pair

Performance comparison of proposed voltage reference circuits

Architecture	$\Delta V_{TH} - V_T$ Core-IO (Measured)	$\Delta V_{TH} - V_T$ Core-Core (Measured)
Technology	180 nm	180 nm
V_{DD} (V)	1	1
I_{DD} (A)	15 μ	1.13 μ
Power Cons. (W)	15 μ	1.13 μ
ΔV_{REF} (V)	± 2.2 m	± 1.225 m
MAPV (%)	0.79	1.43
Temp. Range ($^{\circ}$ C)	25 : 125	25 : 125
Area (mm ²)	0.028	0.0048
Voltage Spread (V)	± 90 m	± 14 m

Conclusion

Voltage reference is integral part of any IC design, therefore understanding it is an essential part of design !!